

GENERAL THETA FOUNDATIONS I: ROBUST COMPATIBILITY OF NUMERICAL WORD REALIZATIONS

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ABSTRACT. We study finite-horizon numerical experiments realized by stochastic registers with prescribed layer widths. For antisymmetric targets, two labels at every command cut admit an exact scalar tensor normal form: deleting affine offsets does not increase the optimal uniform error. A finite parity certificate detects failure of sign compatibility at arbitrary horizon. For constant-magnitude target entries it decides whether two labels can improve on the fair-coin error, and provides a matching constructive alternative. A planar one-command experiment has a complete two-cut error frontier: either cut alone can use two labels at error $\rho/4$, whereas using two labels at both forces error $\rho/2$. The obstruction survives arbitrary intervening hidden states and fixed perturbations. Compatible positive Hankel factors, finite all-word Gram certificates, convergent error bounds, arithmetic width laws and a separately priced finite-bit compiler are retained. The primary resource is nonuniform clocked atomic-row label width; we distinguish actual numerical error from average error and local compatibility certificates. Classical realization and tensor mechanisms are explicitly separated from the finite compatibility conclusions.

1. INTRODUCTION

A word response can have constant signed Hankel rank and constant nonnegative rank at every cut, yet require increasing positive state cardinality when its cut factorizations must use compatible stochastic updates. Positive realization, rather than ordinary linear realization, is therefore the relevant formalism. We study the controlled numerical response

$$(1.1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad b \in \{-1, 1\},$$

for a finite orthogonal alphabet, a unit seed and one selected terminal coordinate query. Our basic resource is nonuniform clocked atomic-row positive-realization label width. The known horizon, row tables and exact real arithmetic are free in that model. A separate theorem below charges them in a uniform finite-bit command-stream model.

The principal new results concern an exact low-width classification, not another numerical test of an inherited asymptotic exponent. Theorem 3.1 identifies the actual optimum for every antisymmetric finite experiment with two labels at each cut. Theorem 3.2 gives a finite parity certificate for one side and a stochastic construction for the other. The certificate can be placed between any two two-label bottlenecks; widths elsewhere are unrestricted. Theorem 3.4 determines an entire one-command error frontier and provides all its rational upper witnesses. In particular the minimum feasible width at separate cuts need not form a jointly feasible profile even at positive error. The target probabilities are uniformly away from zero and one, and Corollary 3.6 retains a strict gap after perturbation.

These statements do not assert that tensor rank classifies arbitrary positive realizations. Their normal form is specific to two labels and antisymmetric target means. The sign dichotomy determines a strict improvement threshold rather than every optimum on its feasible side. Classical scalar stochastic updates and parity elimination are proved as needed and are not claimed as new

algebraic tools. The finite compatibility conclusions concern the prescribed numerical semantics, with every command and the held binary answer explicitly counted.

The finite-horizon question has two logically separate parts: whether a proposed machine realizes every word, and how close the best machine of a prescribed profile can be. Section 5 treats both with exact finite certificates. The all-word Gram identity checks numerical equality rather than average agreement; a backward recursion extracts a failed word when equality does not hold. It yields a polynomial-size existential real formulation of exact feasibility. The even-moment hierarchy and a rational grid theorem then bound the actual optimized terminal error from both sides. Their convergence is a theorem about all legal hidden machines, whereas the current verification program checks given rational realizations, sign systems and finite identities. No globally optimized grid minimum is reported by that program. These tasks must not be conflated.

The ingredients are deliberately distinguished from their use here. Polynomial-time equivalence of probabilistic automata is classical [20]; finite Gram and tensor manipulations are standard in weighted-automata theory [14]. Real quantifier elimination gives effectiveness, not a practical generic minimizer [1]. The categorical-output comparison imports the known nonnegative-rank complexity and field distinction [21]; it is not a hardness theorem for the fixed planar binary-query subclass. The finite certificate formulation adds neither an unpriced state nor an extra query to the numerical experiment.

Three previously established components are retained. First, Section 4 writes the exact finite-horizon object in the language of normalized positive Hankel factorizations and compatible future-response polytopes. This specializes the classical shift-invariant-cone principle; neither positivity nor compatibility is claimed as a new realization concept. For the planar numerical family, ordinary rank, nonnegative rank and normalized positive rank all equal three at every cut. This identifies precisely what separate factorization omits.

Second, write $\Gamma_{\mathcal{A}}(k, B)$ for optimized length- B word quantization loss and $\lambda_{\mathcal{A}}(k, a)$ for the least common dilation of a k -generator polytope containing a times the unit ball. Theorem 7.1 proves the universal comparison

$$(1.2) \quad \sup_{B \geq 1} \frac{-\log(1 - \Gamma_{\mathcal{A}}(k, B))}{B} \leq \log \lambda_{\mathcal{A}}(k, a).$$

It uses arbitrarily many independent packets to remove the initial inradius loss. Thus the paired profiles are related for every finite orthogonal alphabet, including noncommuting alphabets. Equality and optimality among all hidden realizations are not assumed.

Third, Theorem 12.3 gives a table-free finite-bit compiler for finitely many planar angles having fixed linear-space digit procedures. Its parameter search, dyadic probabilities, arithmetic scratch space, horizon counter and finite random words require $O(\log N + \log(1/\varepsilon))$ work bits. Every command is processed once; the output error is uniformly at most a prescribed $\varepsilon > 0$. For the explicit badly approximable algebraic-angle families the work-space order is exactly $\Theta(\log N)$, while the clean stochastic register retains the arithmetic label exponent. This is not an exact fair-bit implementation of arbitrary real rows, and the sharp coefficient of work space is not determined.

The packet, interval and enclosure theorems of the preceding development are reproduced with their proofs in Section 6. The packing-dilation balance there is a conditional meta-corollary, not a classification of finite alphabets. The arithmetic results below are retained: all of their analytic proofs remain in the current paper, with the branch of the finite-type minimum displayed explicitly.

Our arithmetic conclusions go beyond uniform bad approximation. For $\alpha \in \mathbb{R}^r$ with $1, \alpha_1, \dots, \alpha_r$ rationally independent, write

$$(1.3) \quad \omega^*(\alpha) = \limsup_{\|m\|_1 \rightarrow \infty} \frac{-\log \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}}{\log \|m\|_1}.$$

This is the ordinary dual exponent $\omega_{1,r}$ of the row vector α in [17, Definition 1], with an equivalent fixed-dimensional norm; it is not the uniform exponent $\hat{\omega}_{1,r}$ or the simultaneous exponent $\omega_{r,1}$. The alphabet consists of the identity and the rotations through $2\pi\alpha_i$. Throughout, r , the alphabet, $0 < \rho \leq 1/10$ and $0 \leq \varepsilon < \rho/(2\sqrt{2})$ are fixed.

Theorem 1.1 (Minimal Diophantine type). *If $\omega^*(\alpha) = r$, then*

$$(1.4) \quad \lim_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = \frac{r}{2r+1}.$$

For Lebesgue-almost every α , and every fixed $\delta > 0$, there are positive constants c, C , depending on the fixed $\alpha, \delta, r, \rho, \varepsilon$, such that, for $N \geq 2$,

$$(1.5) \quad c \frac{N^{r/(2r+1)}}{(\log(N+2))^{2r(1+\delta)/(2r+1)}} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)} (\log(N+2))^{2r^2(1+\delta)/(2r+1)}.$$

The upper inequalities are achieved by exact common-row machines. No uniformity near rational relations or in growing r is asserted.

The exponent in (1.4) need not imply a two-sided constant-factor law. The earlier dual badly approximable theorem supplies the stronger Θ conclusion on its stated class, and is recovered in Corollary 9.4. Theorem 1.1 instead allows arbitrarily small Diophantine constants along subsequences. The quantitative arithmetic ingredients are classical; the memory conclusions concern their compatible realization and all-hidden-state converse.

There is also a genuine failure of a single exponent, even for one fixed irrational angle.

Theorem 1.2 (Irrational horizons and Liouville fluctuations). *For every irrational α there are horizons $M_n \rightarrow \infty$ such that $W_{M_n,\varepsilon}(\alpha) = \Theta(M_n^{1/3})$. If α is Liouville, then*

$$(1.6) \quad \liminf_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = 0, \quad \frac{1}{3} \leq \limsup_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} \leq \frac{1}{2}.$$

Nevertheless $W_{N,\varepsilon}(\alpha) \rightarrow \infty$. All displayed upper bounds have exact realizations.

The upper limit in (1.6) is not determined here. The result establishes oscillation between subpower subsequences and a positive-power subsequence; it does not replace that distinction by a guessed exponent for all irrationals.

Provenance of the ingredients. The stochastic Hankel normal form and its finite separation dual are normalizations of established realization and convex-separation arguments [12, 16, 14]. Their role is to place the resource, not to claim priority for positive cones. The endpoint mechanism and arithmetic laws are inherited [7, 8, 19]. The distortion-rate comparison and the uniform finite-bit realization with its charged resource ledger were established in the preceding revision. The finite all-word and actual-error framework of Section 5 is retained from the readable revision-46 source. The new statements are the two-label normal form, balanced certificates and robust profile frontiers of Section 3. Dirichlet approximation, continued fractions, metric tail estimates and the algebraic examples remain classical. Section 13 compares the precise objectives and norms, rather than treating adjacent theories as absent.

2. THE NUMERICAL EXPERIMENT AND THE REGISTER

Let $V = \mathbb{R}^D$, let $\mathcal{A}$ be a nonempty finite alphabet, and fix orthogonal matrices U_a , $a \in \mathcal{A}$. Write $U_w = U_{a_N} \cdots U_{a_1}$ in the order of execution. Let $X \subset \mathbb{S}^{D-1}$ be a nonempty finite seed set and $0 < \rho < 1$. The required law is (1.1). Commands are externally supplied, not chosen by a policy inside the simulator. No distribution on command words is part of the specification.

Definition 2.1 (Clocked atomic-row realization). At horizon N , a machine consists of finite sets S_t with $|S_t| = K_t$ for $0 \leq t \leq N$, an initialization probability row E_x on S_0 , stochastic matrices $T_{t,a} : S_{t-1} \rightarrow S_t$, and terminal columns $d_j : S_N \rightarrow [-1, 1]$. Its numerical mean is

$$E_x T_{1,a_1} \cdots T_{N,a_N} d_j.$$

After the atomic query j , it draws and holds the binary answer with that mean. The final answer cut has two labels. Its peak is $\max(2, K_0, \dots, K_N)$. Available labels count even when unreachable; adding unused normalized rows is permitted.

All persistent randomness, input history and protocol distinctions must be in the current label. Only the external epoch, the prescribed horizon and the read-only row tables are free. Table construction, lookup, arithmetic and exact sampling of known real stochastic rows are uncharged. Matrices, rows and state sets may depend on N ; this is nonuniform positive-realization width, not uniform computational space. A common-row construction uses the same command matrices at all command epochs but may still have an N -dependent decoder. It need not answer at earlier stopping lengths.

Let $W_{N,\varepsilon}$ be the least peak among machines whose every binary row differs from (1.1) by at most ε in total variation. In particular $W_{N,\varepsilon} \geq 2$. For fixed finite $K_\bullet$ the optimal actual error $\mathcal{E}_N(K_\bullet)$ is attained: the row parameters lie in a finite product of compact sets and the error is a finite maximum of continuous functions. For planar results we use $X = \{\pm e_1, \pm e_2\}$ and include the identity letter. We keep $\rho \leq 1/10$ and $\varepsilon < \rho/(2\sqrt{2})$.

Lemma 2.2 (Calibration under any external test law). *Fix a seed $x \in X$ and any law on the complete command word, independent of private initialization. Let $Y_t = U_{a_t} \cdots U_{a_1} x$, $Z_t = \mathbb{E}[Y_t | S_t]$ and $q_t = \mathbb{E}\|Z_t\|$. Every machine with row error at most ε satisfies*

$$(2.1) \quad q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon.$$

At a boundary preceding an independent complete packet, the next boundary admits the conditional-mean identity used in Lemma 6.2. No single-step independence inside a packet is asserted.

Proof. For a fixed word, every coordinate of $\mathbb{E}[d(S_N) | w] - \rho Y_N$ has modulus at most 2ε . Since $\|Y_N\| = 1$, averaging gives

$$\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\varepsilon\sqrt{D}.$$

On the other hand this pairing equals $\mathbb{E}\langle Z_N, d(S_N) \rangle$ and is at most $\sqrt{D}q_N$. This proves (2.1). Combining terminal columns is a proof device, not a requirement to provide their joint output distribution. $\square$

3. EXACT TWO-LABEL GEOMETRY AND ROBUST COMPATIBILITY

The positive Hankel normal form below retains the whole future array. For two labels, a smaller exact normal form is available. It gives both a class of finite certificates and an explicit error frontier. The statements in this section concern the actual optimized numerical error, not a sum of local defects or an average over a test law.

3.1. Antisymmetric experiments. We first allow any finite response table, not necessarily one obtained from orthogonal matrices. Let the seeds be paired symbols $X = \{+1, -1, \dots, +n, -n\}$, let $w \in \mathcal{A}^N$, and let $j \in \{1, \dots, d\}$ be the selected query. Suppose the target means satisfy

$$(3.1) \quad f_{-i,w,j} = -f_{i,w,j}, \quad |f_{i,w,j}| \leq 1.$$

The binary law is $(1 + bf_{i,w,j})/2$. Write $\mathcal{E}_2(f)$ for the minimum worst-word binary-TV error of a machine with two available labels at *every* cut $0, \dots, N$. The final held answer also has two labels. Random initialization and arbitrary cut-dependent stochastic matrices are allowed. All extrema below are attained.

Theorem 3.1 (An exact scalar normal form). *For every table satisfying (3.1),*

$$(3.2) \quad 2\mathcal{E}_2(f) = \min_{|h_i| \leq 1, |\beta_{t,a}| \leq 1, |d_j| \leq 1} \max_{i,w,j} \left| f_{i,w,j} - h_i d_j \prod_{t=1}^N \beta_{t,at} \right|.$$

Thus the exact numerical optimization is the distance, in the uniform norm, to the set of bounded decomposable tensors with one factor for the positive seed, one for each command epoch, and one for the query. Every feasible tensor on the right has a two-label stochastic realization. For $N = 0$ the empty product is one.

Proof. Represent a probability row on two labels by its signed coordinate $r = 2p - 1 \in [-1, 1]$. Every stochastic update is an affine map $r \mapsto \alpha + \beta r$ with $|\alpha| + |\beta| \leq 1$; conversely this condition is sufficient. A bounded terminal mean has form $c_j + d_j r$ with $|c_j| + |d_j| \leq 1$. For a word w , the composite update has the form $A_w + B_w r$, where $B_w = \prod_t \beta_{t,at}$. If the two paired initialization coordinates are r_i, r_{-i} , the odd part of the realized mean is therefore

$$\frac{\hat{f}_{i,w,j} - \hat{f}_{-i,w,j}}{2} = h_i B_w d_j, \quad h_i = (r_i - r_{-i})/2.$$

Taking the odd part cannot increase uniform error against the odd target: each resulting residual is half the difference of two original residuals. This operation is realizable without mixing whole machines. Use initialization coordinates $\pm h_i$, replace every update by $r \mapsto \beta_{t,a} r$, and replace the terminal column by $d_j r$. All rows remain stochastic because $|\beta_{t,a}|, |d_j| \leq 1$. This proves that the right side is no larger than twice the original optimum, without retaining a symmetrization coin.

Conversely, initialize the positive label with probability $(1 \pm h_i)/2$, use the stochastic matrix

$$(3.3) \quad \frac{1}{2} \begin{pmatrix} 1 + \beta_{t,a} & 1 - \beta_{t,a} \\ 1 - \beta_{t,a} & 1 + \beta_{t,a} \end{pmatrix},$$

and terminal means $(d_j, -d_j)$. Induction gives the tensor in (3.2). Binary total variation is half the mean error. Compactness of the factor cubes and continuity prove attainment. $\square$

The scalar description of a two-state stochastic matrix is elementary. The content needed here is that deleting all affine offsets produces an admissible optimizer for an odd target, rather than merely a signed linear approximation. No averaging over an unrecorded global random choice is permitted. Tensor descriptions of stochastic processes are classical; Ohta's reachable/null-space reductions concern a different stationary observation model [16]. Equation (3.2) keeps the prescribed command order, bounded query columns and fixed numerical error objective simultaneously.

3.2. Balanced sign certificates. For an entry $e = (i, a_1, \dots, a_N, j)$, form its incidence vector $\iota(e) \in \mathbb{F}_2^M$, where

$$M = n + N|\mathcal{A}| + d.$$

There is one coordinate for each seed factor, each pair (t, a) , and each query factor; $\iota(e)$ marks the factors used by that entry. A set C of nonzero target entries is *balanced* if $\sum_{e \in C} \iota(e) = 0$. Equivalently each factor occurs an even number of times. It is *negative* if $\prod_{e \in C} f_e < 0$.

Theorem 3.2 (A finite sign obstruction and a gap dichotomy). *A negative balanced set C certifies*

$$(3.4) \quad \mathcal{E}_2(f) \geq \frac{1}{2} \min_{e \in C} |f_e|.$$

Suppose in addition that $f_e \in \{0, \rho, -\rho\}$ for a fixed $0 < \rho \leq 1$. Then exactly one of the following holds:

$$(3.5) \quad \begin{aligned} \mathcal{E}_2(f) &= \rho/2, & \text{the nonzero signs do not factor;} \\ \mathcal{E}_2(f) &\leq \rho/4, & \text{the nonzero signs factor.} \end{aligned}$$

Factorability, and hence the alternative in (3.5), is decided by a linear system over $\mathbb{F}_2$. In the first case there is a negative balanced certificate of at most $M + 1$ entries. In the second case the proof constructs a legal two-label machine. Complexity is polynomial in the explicit table size; the table can be exponential in the displayed horizon.

Proof. The product of any decomposable real tensor over a balanced set is nonnegative, since each factor has even exponent. If every tensor entry on C approximated the target within a mean error strictly smaller than $\min_C |f_e|$, it would be nonzero with the target sign. Its product would be negative, a contradiction. Apply Theorem 3.1 to obtain (3.4).

Let $s_e \in \mathbb{F}_2$ encode the nonzero target sign. The signs factor exactly when the system

$$(3.6) \quad \langle \iota(e), z \rangle = s_e \pmod{2} \quad (f_e \neq 0)$$

is consistent. If it is inconsistent, a linear combination of its rows has zero left side and right side one. The corresponding entries are negative and balanced. One may express a contradictory row in a basis of at most M earlier rows, so at most $M + 1$ entries suffice. Inequality (3.4) gives $\mathcal{E}_2(f) \geq \rho/2$; the fair-coin decoder attains the reverse inequality.

If (3.6) has a solution, take the seed and epoch factors to be the prescribed signs ± 1 , and take each query factor to be its prescribed sign times $\rho/2$. Every tensor entry then has magnitude $\rho/2$ and all nonzero targets have the correct sign. Both zero and nonzero entries have mean error at most $\rho/2$. The construction (3.3) realizes these factors with deterministic identity/swap command rows and bounded terminal means. Thus $\mathcal{E}_2(f) \leq \rho/4$. Ordinary finite-field elimination both decides consistency and records the indicated certificate. $\square$

The second alternative need not be an equality: a target which is already a bounded decomposable tensor may have zero error. The theorem classifies the strict-improvement threshold, not every smaller optimum. The certificate is an elementary parity identity, not a new nonnegative-rank theorem or a general polynomial-time minimization algorithm for stochastic automata. The difference is useful: an explicit witness certifies the error against *every* two-label machine, whereas an equivalence test checks a supplied machine.

3.3. Two bottlenecks and an exact finite frontier. Return to the orthogonal experiment. Put

$$(3.7) \quad X = \{\pm e_1, \pm e_2\}, \quad \mathcal{A} = \{I, R\}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad 0 < \rho \leq 1/6.$$

The finite signal range in (3.7) is separate from the smaller sufficient calibration range of the arithmetic theorems. Every target answer probability is in $[(1 - \rho)/2, (1 + \rho)/2]$.

Theorem 3.3 (Two separated bottlenecks). *At any horizon $N \geq 1$, suppose a machine for (3.7) has at most two labels at two distinct cuts $s < t$ in $\{0, \dots, N\}$. Its actual worst-word binary-TV error is at least $\rho/2$, regardless of the widths at all other cuts. The fair-coin output attains this*

bound at every such prescribed profile. The same conclusion holds for any positive signal $\rho \leq 1$, independently of the upper construction used below.

Proof. Freeze all commands before s and after t to I . Between the two cuts retain the two words I^{t-s} and RI^{t-s-1} , with R in the first executed position. Integrating the intermediate computation gives two stochastic maps between the two endpoint alphabets. Pad a one-label endpoint if necessary. The initial probabilities may depend arbitrarily on the seed; the suffix gives one bounded column per query. Thus the odd part of this restricted experiment has the one-step scalar form of Theorem 3.1, even when intermediate alphabets are arbitrarily large.

Four entries of its positive-seed target are

$$f_{1,I,1} = \rho, \quad f_{2,I,2} = \rho, \quad f_{1,R,2} = \rho, \quad f_{2,R,1} = -\rho.$$

Each seed, intervening word and query occurs twice. These four entries form a negative balanced set. Taking the odd part of any putative approximation costs no extra error, so (3.4) gives $\rho/2$. The zero mean decoder proves sharpness. No reset occurs inside the interval and no command is replaced by a free input symbol: collapsing the interval is a converse argument only. $\square$

For $N = 1$ let $\mathcal{E}(k_0, k_1)$ denote the optimum at the two command-boundary widths. A positive minimum is attained even when the allowed profile is infeasible for exact realization.

Theorem 3.4 (A complete two-cut error frontier). *In (3.7) at $N = 1$, the exact frontier is*

$$(3.8) \quad \begin{array}{c|ccc} \mathcal{E}(k_0, k_1) & k_1 = 1 & k_1 = 2 & k_1 \geq 3 \\ \hline k_0 = 1 & \rho/2 & \rho/2 & \rho/2 \\ k_0 = 2 & \rho/2 & \rho/2 & \rho/4 \\ k_0 \geq 3 & \rho/2 & \rho/4 & 0 \end{array}$$

Every upper bound has an explicit rational stochastic realization when ρ is rational. In particular, minimizing either cut separately at tolerance $\rho/4$ permits two labels, but no single machine can attain both minima simultaneously.

Proof. A one-label bottleneck erases all seed dependence; for the identity word, opposite coordinate means force error at least $\rho/2$. Theorem 3.3 treats $(2, 2)$. If either cut has two labels, then, for the identity word, the realized four seed means lie on one affine line $n \cdot y = c$. If its uniform coordinate error is at most δ , then

$$\delta \|n\|_1 \geq \max_{y \in \{\pm \rho e_1, \pm \rho e_2\}} |n \cdot y - c| = \rho \|n\|_\infty + |c| \geq (\rho/2) \|n\|_1.$$

This proves the $\rho/4$ lower bounds after dividing mean error by two. A degenerate segment lies in such a line as well.

Here are matching machines. For $a > 0$, set

$$V_a = \{(a, 0), (0, a), (-a, -a)\}, \quad \beta_a(y) = \frac{1}{3} \left(1 + \frac{2y_1 - y_2}{a}, 1 + \frac{2y_2 - y_1}{a}, 1 - \frac{y_1 + y_2}{a} \right).$$

Whenever $\beta_a(y) \geq 0$, it sums to one and has barycenter y in V_a .

For $(3, 3)$ initialize with $\beta_{2\rho}(\rho x)$. For each vertex $v \in V_{2\rho}$ and command $U \in \{I, R\}$, use the row $\beta_{6\rho}(Uv)$ on $V_{6\rho}$. The six rows are nonnegative: substituting the vertices gives $2(Uv)_1 - (Uv)_2 \geq -6\rho$, the analogous second inequality, and $(Uv)_1 + (Uv)_2 \leq 6\rho$. Use the coordinates of $V_{6\rho}$ as terminal means. They lie in $[-1, 1]$ because $6\rho \leq 1$. The barycenter identity gives exactly ρUx for each word and seed.

For $(2, 3)$ let the two initial labels represent $\pm v$, where $v = (\rho/2, \rho/2)$; initialize e_1, e_2 at $+v$ and their negatives at $-v$. On command U , use the row $\beta_{2\rho}(\pm Uv)$ and decode with $V_{2\rho}$. All four rows are nonnegative. The output is $U[\rho(x_1 + x_2)(1, 1)/2]$, differing from ρUx in each coordinate by at most $\rho/2$ for both I and R .

For $(3, 2)$ initialize with $\beta_{2\rho}(\rho x)$ and decode the two terminal labels as $(1, 1)$ and $(-1, -1)$. For $v \in V_{2\rho}$ use positive-label probability $(1 + \ell(Uv))/2$, where $\ell(y) = (y_1 + y_2)/2$. Here $|\ell(Uv)| \leq 2\rho \leq 1$. The output mean is $\rho((Ux)_1 + (Ux)_2)(1, 1)/2$, again with coordinate error $\rho/2$. All remaining entries in the table follow by padding normalized unused rows, or by the zero decoder. Every coefficient just displayed is rational when ρ is rational. $\square$

For $\rho/4 \leq \varepsilon < \rho/2$, the feasible two-cut profiles are exactly the upward closure of $(2, 3)$ and $(3, 2)$. For $0 \leq \varepsilon < \rho/4$, both command cuts need at least three labels; for $\varepsilon \geq \rho/2$, one at each command cut suffices. The held answer is always charged separately as two labels. These are finite numerical frontiers, not asymptotic claims about an irrational alphabet. They complement rather than replace the growing-width laws.

Theorem 3.5 (An arbitrary-horizon profile frontier). *For the alphabet and seeds in (3.7), let $N \geq 1$ and suppose every command-cut width belongs to $\{1, 2\} \cup \{4, 5, \dots\}$. For any $0 < \rho \leq 1$, the exact optimized error is*

$$(3.9) \quad \mathcal{E}(K_\bullet) = \begin{cases} \rho/2, & \text{some } K_t = 1, \text{ or at least two cuts have } K_t = 2, \\ \rho/4, & \text{exactly one cut has } K_t = 2 \text{ and none has width one,} \\ 0, & \text{every } K_t \geq 4. \end{cases}$$

The constructive machines have deterministic initialization and command rows and rational decoder probabilities when ρ is rational. The excluded width-three profiles are not classified by this statement.

Proof. For a one-label cut freeze every word to the identity; opposite seeds force error $\rho/2$. Two two-label cuts are treated by Theorem 3.3. With one such cut, freeze the entire word to the identity. Its four output means lie in a segment, so the line argument in Theorem 3.4 gives error $\rho/4$.

For the matching construction without a bottleneck, store which of $\{\pm e_1, \pm e_2\}$ is the current vector. The command R permutes these four states, and the decoder is ρ times the vector. If the unique two-label cut is s , track this vector until that cut. At cut s replace the current vector y by $Py = (y_1 + y_2)(1, 1)/2$. There are only two possibilities, $\pm(1, 1)/2$, for these four axis vectors. This is a deterministic compression, with no retained old phase. Thereafter four labels track the quarter-rotation orbit of Py . The first transition out of the bottleneck embeds its two possibilities into these four labels and applies the actual command; subsequent transitions are permutations. At $s = 0$ perform the compression in initialization; at $s = N$ decode the two possibilities directly. The decoder multiplies by ρ . For each axis vector, $\|y - Py\|_\infty = 1/2$; every subsequent quarter rotation preserves this norm. The final mean error is exactly $\rho/2$, proving the second line of (3.9). A fair-coin decoder handles the first line. Available excess labels are padded by normalized unused rows. No random branch or clock history is stored for free. $\square$

Corollary 3.6 (Stability under arbitrary target perturbations). *If a finite binary response table is perturbed by at most η in uniform row TV, its optimal error at any fixed profile changes by at most η . In particular the two-bottleneck lower bound becomes $\max(0, \rho/2 - \eta)$ for any such perturbation of (3.7). The construction at $(2, 3)$ or $(3, 2)$ has error at most $\rho/4 + \eta$. Thus a strict compatibility gap persists whenever $\eta < \rho/8$, including full-support perturbations.*

Proof. For a fixed machine, the two target errors differ by at most η by the triangle inequality; minimizing in both directions proves the first claim. Apply it to (3.8); the gap between $\rho/2 - \eta$ and $\rho/4 + \eta$ is positive exactly when $\eta < \rho/8$. Antisymmetry need not survive the perturbation. $\square$

4. CONTROLLED HANKEL ARRAYS AND COMPATIBLE POSITIVE REALIZATIONS

For a horizon N , define the probability-valued prefix-suffix matrix

$$(4.1) \quad H_t[(x, u), (v, j, b)] = \frac{1 + b\rho\langle e_j, U_v U_u x \rangle}{2}, \quad |u| = t, \quad |v| = N - t.$$

The two columns indexed by $b = \pm 1$ sum to one for every fixed (v, j) . Let $\mathcal{C}_t$ be the product of these binary probability simplices. Thus a point of $\mathcal{C}_t$ is a legal collection of responses to all remaining words and queries, not a joint distribution of those answers. The restriction $L_{t,a} : \mathcal{C}_t \rightarrow \mathcal{C}_{t+1}$ reads the coordinates whose first remaining command is a .

A normalized positive factorization is $H_t = E_t B_t$, where E_t is row stochastic and every row of B_t belongs to $\mathcal{C}_t$. Its least inner dimension is denoted $\text{rank}_{\text{st}} H_t$. The usual nonnegative rank permits arbitrary nonnegative factors and need not impose these simultaneous normalizations. Consequently

$$(4.2) \quad \text{rank } H_t \leq \text{rank}_+ H_t \leq \text{rank}_{\text{st}} H_t \leq K_t$$

for every exact stochastic realization. The last inequality follows by conditioning the remaining experiment on the actual label at cut t . The factor E_t contains its state probabilities; B_t contains its complete future responses, including those of states not reached by a particular prefix.

Proposition 4.1 (Finite-horizon positive Hankel normal form). *For any finite table of binary responses, a profile $K_0, \dots, K_N$ is exactly realizable if and only if there are polytopes $P_t \subset \mathcal{C}_t$, generated by at most K_t points, such that*

$$(4.3) \quad H_0[x, \cdot] \in P_0 \quad (x \in X), \quad L_{t,a} P_t \subset P_{t+1} \quad (0 \leq t < N, a \in \mathcal{A}).$$

The held answer additionally costs two labels. The optimum error for a fixed profile is the least entrywise error from the specified table to one having such a compatible family.

Proof. Given a machine, let $b_{t,s} \in \mathcal{C}_t$ be the conditional response to every remaining word when its current label is s . Its stochastic update gives

$$L_{t,a} b_{t,s} = \sum_{s'} T_{t+1,a}(s, s') b_{t+1,s'}.$$

Taking convex hulls and using the initialization proves necessity. Conversely choose generators $b_{t,s}$, padding by repetitions when necessary. Express each $L_{t,a} b_{t,s}$ as a convex combination of the next generators, and use those coefficients as the stochastic row. Express each root row as a convex combination of the initial generators. At time N the generators themselves give the binary decoder rows. Backward induction on the remaining word proves equality of all responses. The construction is valid on every generator, not only on reachable mixtures. For approximation apply this equivalence to the actual response table of a machine, or to a candidate perturbed table; finite compactness of the stochastic parameter spaces gives attainment. $\square$

This is the finite, controlled, normalized version of the classical shift-invariant-cone realization principle, not a newly discovered characterization of positive systems. In particular, the cone generated by $(1, b)$ for $b \in P_t$ carries the positive restriction maps. Benvenuti-Farina's Theorems 2-3 give the stationary cone formulation; the multiple cuts here are essential [12]. The same compatibility issue was already explicit in the earlier lift formulation [18]. The normal form has exponentially many suffix coordinates. It is exact, but does not make arbitrary width minimization efficient or eliminate hidden directions by projecting them into the observed D -space.

Proposition 4.2 (A finite compatibility certificate). *For proposed generators B_t, B_{t+1} , put $c = L_{t,a}b_{t,s}$ and let $b'_1, \dots, b'_{K_{t+1}}$ be the next generators. The least rowwise uniform defect is*

$$(4.4) \quad \min_{\theta \in \Delta_{K_{t+1}}} \left\| c - \sum_i \theta_i b'_i \right\|_\infty = \max_{\|z\|_1 \leq 1} \left(\langle z, c \rangle - \max_i \langle z, b'_i \rangle \right).$$

It vanishes exactly when that row has a stochastic extension. If root rows are approximated within δ_0 and every cut restriction within δ_t , the resulting machine has worst-word binary-TV error at most $\delta_0 + \sum_{t=1}^N \delta_t$.

Proof. Write the infinity norm as the supremum of its pairing with the unit ℓ_1 ball. The simplex and this ball are compact and convex, so finite bilinear minimax interchanges the extrema. Minimizing the pairing over the simplex gives the maximum over its vertices with the displayed sign. For composition, restriction does not increase the infinity norm and stochastic averaging is a contraction. Induction on the suffix length therefore adds the successive errors. Each binary pair is normalized, so its total variation is exactly either coordinate error. Cancellation may reduce the true error; no equality with the local sum is asserted. $\square$

Proposition 4.3 (Constant separate ranks for the planar experiment). *For the four planar seeds, an identity command, and $0 < \rho \leq 1/10$, at every cut and every horizon,*

$$(4.5) \quad \text{rank } H_t = \text{rank}_+ H_t = \text{rank}_{\text{st}} H_t = 3.$$

Each factorization embeds in a complete machine when all other cuts are unrestricted. Nevertheless the compatible peak can grow as in Theorems 1.1 and 1.2.

Proof. The entries in (4.1) are affine functions of the two coordinates of $\rho U_u x$, so ordinary rank is at most three. Identity prefixes, identity suffixes, both coordinate queries and the four seeds give a rank-three submatrix. For the upper factorization, an equilateral triangle of circumradius 2ρ contains the disk of radius ρ . Encode $\rho U_u x$ in its three vertices v_i by barycentric coordinates. The continuation row at v_i is $(1 + b\langle e_j, U_v v_i \rangle)/2$, which is legal since $\|v_i\| = 2\rho < 1$. This is a normalized three-factor decomposition and proves the equalities. To embed it, retain the exact history until this cut, draw its factor label, retain that label and the remaining history afterwards, then use the factor decoder. Both retained histories are finite at a fixed horizon. They are not free registers in a simultaneous three-state realization. The growth assertions follow from the stated all-hidden converses, not from a claimed new separation of signed and positive realization in classical stationary theory. $\square$

5. FINITE CERTIFICATES FOR THE WHOLE NUMERICAL EXPERIMENT

The Hankel normal form uses all suffixes. We now give an equivalent finite certificate whose size is polynomial in a *displayed* horizon and proposed widths. We then treat the actual optimal error, not the sum of local compatibility defects. Throughout this section X and $\mathcal{A}$ are nonempty, $|\mathcal{A}| = m$, $D \geq 1$, and $K_0, \dots, K_N \geq 1$. The held binary answer is a separate two-label cut. A horizon is supplied in unary for statements about input size. Rational numbers have binary encodings; a real algebraic coefficient is supplied by a polynomial and a rational isolating interval. These conventions do not give an oracle for arbitrary real numbers.

In this section $T_{t,a}$ labels the edge from cut t to cut $t+1$, a reindexing of Definition 2.1. Write E_x for an initialization row, $T_{t,a}$ for the K_t -by- K_{t+1} row-stochastic transition, and $d_j \in [-1, 1]^{K_N}$ for the terminal mean column. Define the residual mean

$$(5.1) \quad R_{x,w,j} = E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j - \rho \langle e_j, U_{a_N} \cdots U_{a_1} x \rangle.$$

Thus the actual worst-word binary-TV error is $e(\mathcal{M}) = \frac{1}{2} \max_{x,w,j} |R_{x,w,j}|$. In particular, centering the numerical response does not turn a bounded stochastic decoder into an unrestricted signed readout.

Set $n_t = K_t + D$ and form rectangular block matrices and vectors

$$(5.2) \quad A_{t,a} = \text{diag}(T_{t,a}, U_a^\top), \quad u_x = (E_x, -\rho x^\top), \quad v_j = \begin{pmatrix} d_j \\ e_j \end{pmatrix}.$$

The order and transpose in this definition are essential for noncommuting commands. Equation (5.1) equals $u_x A_w v_j$.

Theorem 5.1 (A finite all-word certificate). *Define*

$$(5.3) \quad G_0 = \sum_{x \in X} u_x^\top u_x, \quad G_{t+1} = \sum_{a \in \mathcal{A}} A_{t,a}^\top G_t A_{t,a}, \quad S_2 = \sum_{j=1}^D v_j^\top G_N v_j.$$

Then

$$S_2 = \sum_{x \in X} \sum_{w \in \mathcal{A}^N} \sum_{j=1}^D R_{x,w,j}^2.$$

Consequently a legal proposed machine is exact on every word if and only if $S_2 = 0$. For rational data this can be checked by exact arithmetic in polynomial time in the displayed data. If $S_2 > 0$, a seed, word and query with nonzero residual can also be found in polynomial time, without listing all words.

Exact feasibility of a prescribed profile for an algebraic orthogonal-word target has an existential real formula of polynomial size in $N, m, D, |X|$ and the explicitly displayed widths. Its equations may all be taken to have degree at most three after auxiliary arithmetic variables are introduced. Feasibility has a real algebraic witness whenever it has a real witness.

Proof. Induction in (5.3) gives $G_t = \sum_{x, |u|=t} (u_x A_u)^\top (u_x A_u)$. Pairing with v_j proves the identity. Every summand is a real square, so zero means equality of all numerical responses, not merely equality on average. No assumption about the ranks of reachable distributions or the observability of individual states is used.

For a witness, compute backwards

$$C_N = \sum_j v_j v_j^\top, \quad C_t = \sum_a A_{t,a} C_{t+1} A_{t,a}^\top.$$

Select x with $u_x C_0 u_x^\top > 0$. Given a current row u at cut t with $u C_t u^\top > 0$, at least one command satisfies $u A_{t,a} C_{t+1} A_{t,a}^\top u^\top > 0$. Append it and replace u by $u A_{t,a}$. At the last cut choose j with $u v_j \neq 0$. The matrices are positive semidefinite by their recursions, so this procedure cannot encounter a dead end. There are $O(Nm)$ matrix operations at dimensions at most $\max_t n_t$. For rational input, common-denominator bounds on these finite matrix products also give polynomial intermediate bit length. Exact algebraic sign testing gives the corresponding effective statement for algebraic inputs; no unspecified real-number comparison is invoked.

For existence, treat E, T, d, G as unknowns. Impose nonnegativity and unit row sums on E, T , the bounds on d , the recursions (5.3), and $S_2 = 0$. Each recursion entry has degree at most three, and its number of terms is polynomial in the matrix sizes. The Gram variables need no independent positivity constraint: the initial condition and recurrence already force the displayed sum of squares. If target coefficients are algebraic rather than rational, include their defining polynomials and isolating inequalities. Arithmetic circuit variables reduce larger degrees to quadratic equations. Elimination over real closed fields decides existence and permits algebraic sample points. This last conclusion is the usual real-algebraic sampling principle [1], not a polynomial-time optimizer. $\square$

The linear-algebraic equivalence mechanism is classical: compare Tzeng's equivalence algorithm [20] and the weighted-automata Hankel and Gram constructions of Balle–Panangaden–Precup [14]. The assertion here makes the finite clock, stochastic constraints, all seed initializations, and bounded query columns simultaneous. It must not be read as a new discovery of polynomial-time weighted-automata equivalence. Also, $S_2 \approx 0$ in floating point is *not* the exact certificate $S_2 = 0$.

5.1. A hierarchy for the actual optimal error. Let $Q = |X|Dm^N$. For each even integer $p \geq 2$, define

$$(5.4) \quad L_p(\mathcal{M}) = \frac{1}{2} \left(Q^{-1} \sum_{x,w,j} R_{x,w,j}^p \right)^{1/p}, \quad \ell_p(K_\bullet) = \min_{\mathcal{M}} L_p(\mathcal{M}),$$

where every minimization uses the *same* product of stochastic row simplices and bounded decoder cubes for the specified profile. Put $\mathcal{E}(K_\bullet) = \min_{\mathcal{M}} e(\mathcal{M})$.

Theorem 5.2 (Moment certificates and the error frontier). *All the preceding minima are attained. For every even $p \geq 2$,*

$$(5.5) \quad \ell_p(K_\bullet) \leq \mathcal{E}(K_\bullet) \leq Q^{1/p} \ell_p(K_\bullet).$$

The numbers ℓ_p increase along the even integers and converge to $\mathcal{E}$. The running minima of the right sides of (5.5) decrease to the same value. In particular, $\ell_2 = 0$ if and only if the profile admits an exact realization. For each fixed p , its moment sum is computable by a finite homogeneous polynomial recurrence without enumerating words. For fixed $\max_t(K_t + D)$, this recurrence has size polynomial in $p, N, m, D, |X|$.

For algebraic target data the actual optimum $\mathcal{E}$ and a minimizing machine are algebraic and effectively obtainable. For a finite prescribed width cap, the resulting threshold values give the exact finite-horizon error/width frontier. This is an effective finite optimization statement, not a closed formula or a polynomial-time algorithm for growing width.

Proof. The parameter set is nonempty and compact, and the objectives are continuous. The inequalities for a vector of Q numbers are

$$\left(Q^{-1} \sum |r_i|^p \right)^{1/p} \leq \max_i |r_i| \leq Q^{1/p} \left(Q^{-1} \sum |r_i|^p \right)^{1/p}.$$

The first inequality survives minimization. For the second, use a minimizer of L_p as a candidate for $\mathcal{E}$; no exchange of a maximum over words with a minimum over machines is made. Normalized finite-measure L^p norms increase with p , as do their minima. Since $Q^{1/p} \rightarrow 1$, the two sequences converge as asserted, including the case of a zero optimum.

The recurrence is particularly transparent in polynomial form. Let z be a row variable of the appropriate dimension and set

$$(5.6) \quad F_N(z) = \sum_j (zv_j)^p, \quad F_t(z) = \sum_a F_{t+1}(zA_{t,a}).$$

Then $\sum_x F_0(u_x)$ is the moment sum. A homogeneous degree- p polynomial in n_t variables has $\binom{n_t+p-1}{p}$ coefficients. Repeated polynomial multiplication and linear substitution compute these coefficients in polynomially many arithmetic operations when n_t is fixed. This is the symmetric-power form of the familiar tensor product of weighted representations. It does not imply a bound polynomial in both p and unbounded n_t .

For the last assertion one may instead introduce all finitely many word inequalities $-2e \leq R_{x,w,j} \leq 2e$ and minimize $e \in [0, 1]$. Projection of this compact semialgebraic feasible set gives its semialgebraic attainable-error set. Its left endpoint is algebraic, and real-algebraic sampling gives a machine above that endpoint. The horizon and word set are finite; the number of those

inequalities can be exponential. Standard real quantifier elimination proves the claim, with no missing infinite-horizon limit. Apply it successively to the finitely many allowed integer profiles or to padded constant caps. The held two-label output cut is then included in the reported peak. $\square$

For a *given* machine and tolerance factor $1 + \delta$, choosing an even $p \geq \log Q / \log(1 + \delta)$ gives lower and upper error certificates with ratio at most $1 + \delta$. This statement concerns checking the supplied machine, not finding a global moment minimizer. A certificate for a globally optimized bound must certify that optimization as well. A numerical local minimum has no such status. For fixed dimensions, exact rational coefficient growth in (5.6) is bounded by polynomially many input bit lengths, products and multinomial factors; certified root intervals can replace irrational p th roots in the output. The implementation below uses exact degree-two and degree-four recurrences, not an unreported large-degree optimization.

5.2. Finite precision without changing boundary widths. For an integer $L \geq 1$, let $\mathcal{G}_L$ contain all machines whose initial and transition probabilities, and positive-answer probabilities, are integral multiples of $1/L$. Let $e_L = \min_{\mathcal{M} \in \mathcal{G}_L} e(\mathcal{M})$ and set

$$(5.7) \quad C(K_\bullet) = (K_0 - 1) + \sum_{t=1}^N (K_t - 1) + 1.$$

These finite sets are nonempty even when some cuts have only one label.

Theorem 5.3 (A finite two-sided rational certificate). *For every finite binary target, without any orthogonality assumption,*

$$(5.8) \quad \max\{0, e_L - C(K_\bullet)/L\} \leq \mathcal{E}(K_\bullet) \leq e_L.$$

Consequently dyadic machines with the same available boundary profile attain every strictly larger error than $\mathcal{E}$. The theorem does not assert a dyadic or rational exact optimizer at the boundary. If two target probability tables differ by at most η uniformly, their optimal errors at the same profile differ by at most η . Thus a strict positive feasibility or infeasibility margin persists under sufficiently small target perturbations.

Proof. Round down the first $k - 1$ coordinates of any probability row of size k to multiples of $1/L$, and give the remaining mass to the last coordinate. The row stays stochastic and its TV change is at most $(k - 1)/L$. Round a terminal positive-answer probability to within $1/L$. Couple the two machines at initialization and then at each common command, or telescope the products by TV contraction. For every word the total output change is bounded by (5.7) divided by L . Applying this construction to a minimizing real machine gives $e_L \leq \mathcal{E} + C/L$; inclusion of the grid in the full parameter set gives $\mathcal{E} \leq e_L$. Both use the same labels; no scratch configuration is hidden in that assertion. Choosing $L = 2^b$ sufficiently large proves the strict-margin assertion. Finally, for any fixed machine its errors against two targets differ by at most η by the triangle inequality. Minimize in both directions. $\square$

An exhaustive grid search is a terminating certificate algorithm at fixed L , not an efficient proposal for large instances. If the target is supplied by certified intervals of radius η , the two endpoints in (5.8) can each be enlarged by η . Dyadic row sampling uses b fresh fair bits at denominator 2^b ; index, table and scratch storage still have to be charged in an implementation. The separate uniform compiler below supplies its own workspace accounting. The rational approximation statement must not be strengthened to exact rational factorization: such field restrictions genuinely matter in nonnegative realization [21].

Proposition 5.4 (An exact calibrated finite frontier). *Take $N = 0$, the four planar seeds $\{\pm e_1, \pm e_2\}$, two coordinate queries and $0 < \rho \leq 1/2$ (this wider signal range is used only for the*

present finite example). If k is the available initialization alphabet, the optimal binary-TV errors are

$$(5.9) \quad \mathcal{E}(1) = \rho/2, \quad \mathcal{E}(2) = \rho/4, \quad \mathcal{E}(k) = 0 \quad (k \geq 3).$$

The same formula holds for an arbitrary horizon consisting only of identity commands, with $k = \min_t K_t$. In all cases the final held answer still costs two labels.

Proof. A one-state decoder is one point of the mean square. Approximating $\pm \rho e_1, \pm \rho e_2$ by a common point has minimum uniform coordinate error ρ , attained at zero. For two states every seed mean lies on one affine line $n \cdot y = c$, with $n \neq 0$; a degenerate segment may be embedded in any such line. If all coordinate errors are at most d , duality of ℓ_∞ and ℓ_1 gives

$$d\|n\|_1 \geq \max_{y \in \{\pm \rho e_1, \pm \rho e_2\}} |n \cdot y - c| = \rho\|n\|_\infty + |c| \geq \frac{\rho}{2}\|n\|_1.$$

The diagonal segment with endpoints $\pm(\rho/2, \rho/2)$ attains coordinate error $\rho/2$: encode e_1, e_2 at its positive endpoint and their negatives at its negative endpoint. Divide mean errors by two to obtain binary TV. Three legal mean vectors $(2\rho, 0), (-\rho, 2\rho), (-\rho, -2\rho)$ enclose all four targets, so their barycentric initialization gives exact responses. For identity-only words, condition at a narrowest cut to obtain the same one-cut lower bound. Initialize the appropriate construction and carry its label through all cuts, padding unused labels as necessary, to obtain the upper bound. $\square$

5.3. A complexity boundary with a different output convention. The following comparison concerns one *categorical* terminal output, not a selected member of a family of binary queries. Its width is the pre-output memory alphabet; counting the held output would separately add its alphabet size.

Proposition 5.5 (Normalized factorization and the real field). *Given an explicitly listed rational row-stochastic matrix P , deciding whether it has a stochastic factorization $P = ED$ of inner dimension at most k is $\exists\mathbb{R}$ -complete. It is already a zero-command numerical realization problem. Rational target data need not have a rational exact optimizer at the minimum real inner dimension.*

Proof. Membership follows from the nonnegative variables, row-sum equations and quadratic equations $P = ED$. For hardness, remove the zero rows of a nonzero nonnegative rational matrix A and divide each remaining row by its positive row sum, giving P . Positive diagonal row scaling and removal of zero rows preserve nonnegative rank. If $P = WH$ is a nonnegative factorization, put $h_i = \sum_j H_{ij}$. For $h_i > 0$ set $D_{ij} = H_{ij}/h_i$ and $E_{xi} = W_{xi}h_i$. If $h_i = 0$, replace that unused decoder row by any probability row and set its E column to zero. Then $D\mathbf{1} = \mathbf{1}$ and $E\mathbf{1} = P\mathbf{1} = \mathbf{1}$. Thus stochastic inner dimension is exactly nonnegative rank, over either the real or rational field. Shitov's Theorem 2 gives $\exists\mathbb{R}$ -hardness and his Corollary 3 gives rational matrices with different minimum ranks over the rational and real algebraic fields [21]. Row normalization transfers both statements as just shown. Zero input matrices can be treated separately and do not affect the reduction. $\square$

This transfers a known complexity theorem; it is not a new hardness proof for nonnegative rank. Nor does it establish the same hardness for the fixed-dimensional orthogonal binary-query subclass. In that subclass Theorem 5.1 supplies the compact exact feasibility formula, while the inherited arithmetic theorems determine specific asymptotic widths. These are different assertions.

6. FINITE WORD PROFILES AND COMPATIBLE ENCLOSURES

The results in this section apply to every finite orthogonal alphabet, including noncommuting alphabets. No connectedness, arithmetic condition or spectral gap is assumed. Invariant directions may make a converse profile zero; this is a property of the stated profile, not an omitted hypothesis.

6.1. An optimized distortion. For $B, k \geq 1$ and a probability vector p on the actual words $\mathcal{A}^B$, define

$$(6.1) \quad D_k(p) = 1 - \max_{\substack{u \in \mathbb{S}^{D-1} \\ v_1, \dots, v_k \in \mathbb{S}^{D-1}}} \sum_{w \in \mathcal{A}^B} p_w \max_{1 \leq j \leq k} \langle v_j, U_w u \rangle,$$

$$(6.2) \quad \Gamma_{\mathcal{A}}(k, B) = \max_{p \in \Delta(\mathcal{A}^B)} D_k(p), \quad \chi_{\mathcal{A}}(k, B) = -\log(1 - \Gamma_{\mathcal{A}}(k, B)).$$

We allow $\chi = +\infty$. Repeated product matrices can be merged for the optimization: their probability masses add. Their representative words must still be executed one ordinary command at a time.

Proposition 6.1 (Finite-data properties). *The maxima in (6.1)–(6.2) are attained, $0 \leq \Gamma_{\mathcal{A}}(k, B) \leq 1$, and the profile is invariant under simultaneous orthogonal changes of coordinates. It is nonincreasing in k and unchanged by duplicating a letter. For two indexed orthogonal alphabets U, V ,*

$$(6.3) \quad |\Gamma_U(k, B) - \Gamma_V(k, B)| \leq B \max_a \|U_a - V_a\|_{\text{op}}.$$

For a fully listed finite alphabet of real-algebraic matrices and fixed integers k, B, D , $\Gamma_{\mathcal{A}}(k, B)$ and an optimizing word law are real-algebraic and can be determined by real quantifier elimination.

Proof. All parameter spaces are compact and the objective is continuous. Its maximum over directions is convex in p , so D_k is continuous and concave. The maximum scalar product is at most one. By choosing a single direction parallel to $\sum p_w U_w u$, the inner maximum is nonnegative; hence $D_k \in [0, 1]$. Conjugating all directions proves invariance. Additional centers only increase that inner maximum. Product telescoping gives $\|U_w - V_w\|_{\text{op}} \leq B \max_a \|U_a - V_a\|_{\text{op}}$, and maxima and minima preserve this uniform bound.

For the last assertion, enumerate the product matrices. Unit-sphere equations, simplex inequalities and finitely many maximum operations define a compact semialgebraic optimization with algebraic coefficients. Maximum operations can be expressed by finite alternatives or auxiliary variables. Quantifier elimination over the real closed field of real-algebraic numbers determines the endpoint of the feasible value interval and a witness [1]. The word table may have $|\mathcal{A}|^B$ entries; this is not a polynomial-time assertion or a decision procedure for an infinite spectral gap. No logarithm needs to be eliminated: finite budget comparisons can instead use products of $1 - \Gamma$. $\square$

Lemma 6.2 (Endpoint contraction; inherited mechanism). *Let a complete random word W with law p be independent of all information before its start. Internal letters of W may be correlated. If the endpoint register has at most k labels, then*

$$(6.4) \quad q' \leq (1 - D_k(p))q.$$

There is no bound on the intermediate registers.

Proof. Write $z_s = \mathbb{E}[Y \mid S = s]$ at the start. Composing all internal stochastic rows yields a common endpoint kernel $T(j \mid s, w)$. Since the entire word and the machine's fresh randomness are independent of the past,

$$z'_j = \mathbb{E}[U_W z_S \mid S' = j].$$

This is an endpoint identity; it would not generally hold after conditioning on a correlated next letter inside the packet. Choose $v_j = z'_j / \|z'_j\|$ for nonzero z'_j , and any unit vector otherwise. Then

$$\begin{aligned} q' &= \sum_s \mathbb{P}(S = s) \sum_w p_w \sum_j T(j \mid s, w) \langle v_j, U_w z_s \rangle \\ &\leq \sum_s \mathbb{P}(S = s) \|z_s\| \max_{\|u\|=1} \sum_w p_w \max_j \langle v_j, U_w u \rangle \leq (1 - D_k(p))q. \end{aligned}$$

Terms with $z_s = 0$ contribute zero. Replacing an executable assignment by the largest pairing is a relaxation in the correct direction. Arbitrary unobservable hidden coordinates are included in s ; no statewise observable-vector hypothesis is used. This is the conditional-centroid mechanism of [7, 8], now applied to the optimized finite word datum. $\square$

For an available profile $K_\bullet$, let $I_0 = 0$ and define

$$(6.5) \quad I_t = \max \left\{ I_{t-1}, \max_{1 \leq B \leq t} [I_{t-B} + \chi_{\mathcal{A}}(K_t, B)] \right\}, \quad \mathcal{I}_N(K_\bullet) = I_N.$$

If $\Gamma_{\mathcal{A}}(K_t, B) = 1$, the interval annihilates the conditional-centroid amplitude and is incompatible with any positive terminal calibration. This is the meaning of an infinite charge. The recurrence is the largest sum of charges of disjoint command intervals, with an interval of length B ending at t charged by $\chi_{\mathcal{A}}(K_t, B)$. The recurrence depends only on the alphabet and the available profile.

Theorem 6.3 (A nonuniform interval budget). *Every machine with $\kappa = \rho/\sqrt{D} - 2\varepsilon > 0$ satisfies*

$$(6.6) \quad \mathcal{I}_N(K_\bullet) \leq \log(1/\kappa).$$

More generally its optimal actual terminal error obeys

$$(6.7) \quad \mathcal{E}_N(K_\bullet) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - e^{-\mathcal{I}_N(K_\bullet)} \right]_+.$$

For a uniform peak bound K , every $B \leq N$ gives

$$(6.8) \quad \lfloor N/B \rfloor \chi_{\mathcal{A}}(K, B) \leq \log(1/\kappa).$$

For a fixed B, k with $\Gamma_{\mathcal{A}}(k, B) > 0$,

$$(6.9) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B - 1 + \frac{B \log(1/\kappa)}{\chi_{\mathcal{A}}(k, B)}.$$

If $\chi = +\infty$, the last fraction is zero.

Proof. Choose any maximizing disjoint interval family in (6.5). On each selected interval put an independently sampled optimizing complete word from Proposition 6.1. All gaps and the final remainder contain fixed deterministic command words. A deterministic orthogonal command followed by a stochastic update cannot increase the conditional-mean norm, by conditional Jensen. Lemma 6.2 therefore gives

$$q_N \leq \prod_{(t-B, t]} (1 - \Gamma_{\mathcal{A}}(K_t, B)) = e^{-\mathcal{I}_N(K_\bullet)},$$

since $q_0 = 1$. The chosen law can depend on the available layer sizes, not on private states. Wordwise correctness requires the same machine to pass it. Lemma 2.2 proves (6.6) and (6.7); taking consecutive intervals proves (6.8).

For (6.9), discard candidate endpoints before B . Select the earliest remaining endpoint, delete all later candidates less than B away, and repeat. If M is the original number and J the number selected, the explicit count is

$$M - (B - 1) \leq BJ.$$

The associated length- B intervals are disjoint and each contributes at least $\chi_{\mathcal{A}}(k, B)$. Applying (6.6) gives the result. No restriction on the other registers was introduced. $\square$

The cap union estimate below is the geometric certificate already used in the predecessor packet method [7]. The finite incidence formulation specifies how to optimize that certificate on a fully listed word set.

Proposition 6.4 (Uniform cap certificates). *Suppose a law p on length- B words satisfies*

$$\sup_{u,v \in \mathbb{S}^{D-1}} p\{w : \|U_w u - v\| < \delta\} \leq b.$$

Then $\Gamma_{\mathcal{A}}(k, B) \geq \frac{1}{2}\delta^2(1 - kb)_+$. For algebraic product matrices and algebraic δ , the best such b is the value of a finite linear program once cap incidence types have been decided by real quantifier elimination.

Proof. The union of k open caps has probability at most kb . Outside it, $1 - \max_j \langle v_j, U_w u \rangle = \frac{1}{2} \min_j \|U_w u - v_j\|^2 \geq \delta^2/2$. For the algorithmic assertion, a subset of the finite word list either can or cannot be contained in one such cap for some u, v . This is a finite system of algebraic inequalities, including the sphere equations. For every feasible subset A impose $\sum_{w \in A} p_w \leq b$, along with the simplex constraints, and minimize b . Every actual incidence set is feasible; conversely a feasible subset is included in an actual incidence set and has no greater mass. Thus these constraints give exactly the desired supremum. The exponentially large enumeration is not concealed. $\square$

6.2. A separate exact-synthesis profile. Fix $0 < a < 1$. For k such that the following family is nonempty, let

$$(6.10) \quad \lambda_{\mathcal{A}}(k, a) = \min_P \inf \{\lambda \geq 1 : U_b P \subseteq \lambda P \text{ for all } b \in \mathcal{A}\},$$

where P ranges over convex hulls of at most k points in $\mathbb{B}^D$ containing $a\mathbb{B}^D$. An empty family has value $+\infty$.

Theorem 6.5 (Compatible enclosure synthesis). *The finite value in (6.10) is attained and lies in $[1, a^{-1}]$. If $\rho\lambda_{\mathcal{A}}(k, a)^N \leq a$, there is an exact machine with $\max(k, 2)$ labels, common command matrices and a horizon-dependent decoder. Each row has at most $D + 1$ successors. Algebraic matrices and algebraic a give an algebraically decidable profile value and algebraic row witnesses. For two indexed alphabets with a common feasible family,*

$$(6.11) \quad |\lambda_U(k, a) - \lambda_V(k, a)| \leq a^{-1} \max_b \|U_b - V_b\|_{\text{op}}.$$

Proof. Parameterize P by k points in the closed unit ball, allowing repetition. Containment of $a\mathbb{B}^D$ is closed in these coordinates. Every such P satisfies $U_b P \subseteq \mathbb{B}^D \subseteq a^{-1}P$. The containment conditions are closed on this compact set and $1 \leq \lambda \leq a^{-1}$; hence a minimum exists.

Let its vertices be v_s and λ its dilation. Initialize with a convex representation of ax . For each b, s , represent $\lambda^{-1}U_b v_s$ by the same vertices and use its convex coefficients as the row $T_b(s, \cdot)$. Induction gives

$$\mathbb{E}[v_{S_t} \mid w] = a\lambda^{-t}U_w x.$$

The decoder $d_j(s) = \rho\lambda^N \langle e_j, v_s \rangle / a$ lies in $[-1, 1]$ and gives (1.1). These are actual rows throughout the run, not independently chosen checkpoint factorizations. Carathéodory gives $D + 1$ successors; there are $|\mathcal{A}|k$ command rows, and their construction and storage remain uncharged.

The defining finite problem is semialgebraic: use convex coefficients for transformed vertices and a universal ball containment condition, then eliminate quantifiers. To prove (6.11), set $e = \max_b \|U_b - V_b\|_{\text{op}}$. For every feasible P , $V_b P \subseteq U_b P + e\mathbb{B}^D \subseteq (\lambda + e/a)P$. Interchanging U, V completes the proof. $\square$

The two profiles have different quantifiers. The converse tests every hidden machine; the enclosure theorem builds a particular positive realization. Neither theorem asserts that $\lambda_{\mathcal{A}}$ is the optimum over hidden machines. This is why finite-scale lower and upper bounds can be used even when they do not match.

Proposition 6.6 (Bounded word recoding). *Suppose every command of $\mathcal{A}$ is a product of exactly L commands of an alphabet $\mathcal{B}$ on the same space, with the same seeds and queries. Then*

$$\Gamma_{\mathcal{A}}(k, B) \leq \Gamma_{\mathcal{B}}(k, LB), \quad \lambda_{\mathcal{A}}(k, a) \leq \lambda_{\mathcal{B}}(k, a)^L, \quad W_{N, \varepsilon}(\mathcal{A}) \leq W_{LN, \varepsilon}(\mathcal{B}).$$

An identity letter permits words of length at most L to be padded. If both alphabets contain an identity and admit such bounded recodings into the other, the existence and value of a logarithmic width exponent agree.

Proof. Push forward a packet law along the word substitution to obtain the first inequality. Iterate the enclosure containments for the second. For the last inequality, compose the L stochastic rows of a $\mathcal{B}$ machine into one row for the corresponding $\mathcal{A}$ command, retaining only the block-end label. All rows may depend on the external epoch. This is an implementation of the atomic $\mathcal{A}$ interface, not a claim that its physical letters cost L letters. Applying the two comparisons and monotonicity under final identity padding proves the exponent statement. $\square$

Corollary 6.7 (Packing–dilation balance). *Fix a calibrated experiment, an inradius $a > \rho$, and positive numbers u, v . Suppose that for every sufficiently large k there are a length $B(k) \leq Ck^u$ and a word law for which*

$$\sup_{x, z \in \mathbb{S}^{D-1}} p\{w : \|U_w x - z\| < ck^{-v}\} \leq (2k)^{-1}.$$

Suppose also that $\lambda_{\mathcal{A}}(k, a) \leq \exp(C'k^{-u-2v})$. Then

$$W_{N, \varepsilon} = \Theta(N^{1/(u+2v)}).$$

The same conclusion holds with $u = 0$ when the lengths are uniformly bounded.

Proof. Proposition 6.4 gives $\Gamma(k, B(k)) \geq c^2 k^{-2v}/4$. Equation (6.8) implies $N \leq C_1 k^{u+2v}$ for a feasible peak k . Conversely choose k at a sufficiently large constant times $N^{1/(u+2v)}$ so that $C'Nk^{-u-2v} \leq \log(a/\rho)$, and apply Theorem 6.5. The hypotheses concern finite word orbits and compatible convex containments, and impose no commutativity on the matrices. $\square$

7. A DISTORTION–DILATION COMPARISON

The lower and upper profiles defined in Section 6 do not have identical quantifiers. They nevertheless obey a universal inequality. Define the optimized per-command distortion rate

$$(7.1) \quad \mathfrak{r}_{\mathcal{A}}(k) = \sup_{B \geq 1} \frac{-\log(1 - \Gamma_{\mathcal{A}}(k, B))}{B}.$$

All logarithms in this paper are natural. The value $+\infty$ has its usual extended meaning.

Theorem 7.1 (Distortion bounds every common enclosure). *For every finite orthogonal alphabet and every feasible k, a in (6.10),*

$$(7.2) \quad \mathfrak{r}_{\mathcal{A}}(k) \leq \log \lambda_{\mathcal{A}}(k, a).$$

In particular, every finite-word certificate gives the lower enclosure bound $\lambda_{\mathcal{A}}(k, a) \geq (1 - \Gamma_{\mathcal{A}}(k, B))^{-1/B}$. The assertion does not require commuting commands, a spectral gap, or any Diophantine hypothesis.

Proof. Take any feasible polytope $P = \text{conv}(v_1, \dots, v_k)$, contained in the unit ball and containing a times that ball, with common dilation λ . Use the stochastic rows representing $\lambda^{-1}U_b v_s$ by the same vertices. Fix a unit seed u and initialize the mean at au . This construction makes sense at every length, without imposing a terminal decoder. For each complete word w of length n it satisfies

$$\mathbb{E}[v_{S_n} \mid w] = a\lambda^{-n}U_w u.$$

Let $Y_n = U_w u$ and $q_n = \mathbb{E} \|\mathbb{E}[Y_n | S_n]\|$. Pairing with the bounded vector v_{S_n} gives, under every word law,

$$(7.3) \quad q_n \geq \mathbb{E} \langle Y_n, v_{S_n} \rangle = a\lambda^{-n}.$$

Fix B and independently repeat an optimizing length- B word law m times. The packet lemma applies at the m endpoints, each of which has at most k labels. Since $q_0 = 1$,

$$a\lambda^{-mB} \leq q_{mB} \leq (1 - \Gamma_{\mathcal{A}}(k, B))^m.$$

Taking m th roots and letting $m \rightarrow \infty$ removes the inradius factor. Hence $\lambda^{-B} \leq 1 - \Gamma_{\mathcal{A}}(k, B)$. Take the supremum over B and then the minimum over enclosures. In particular, $\Gamma = 1$ is incompatible with a finite feasible enclosure at this k , so no undefined logarithmic operation was used. $\square$

The removal of the initial inradius loss by repetition is important: a single-horizon application would leave an additive $\log(1/a)$. Theorem 7.1 is a comparison, not a duality asserting equality or a characterization of arbitrary hidden width. The Hankel normal form in Proposition 4.1 permits general future-response polytopes, whereas (6.10) uses one common polytope in the observed vector space.

Corollary 7.2 (Nonuniform enclosure chains). *Suppose $P_t \subset \mathbb{B}^D$ has at most K_t generators, $a\mathbb{B}^D \subset P_0$, and $U_b P_{t-1} \subset \lambda_t P_t$ for every command. Then*

$$\mathcal{I}_N(K_\bullet) \leq \log(1/a) + \sum_{t=1}^N \log \lambda_t.$$

Proof. Use the cut-dependent convex rows and the initialization with mean au . The same induction gives mean $a(\prod_t \lambda_t)^{-1} U_w u$. Choose the independent packet law attaining the interval recurrence. The pairing argument bounds the final amplitude below by this factor, while the packet theorem bounds it above by $\exp(-\mathcal{I}_N(K_\bullet))$. Taking logarithms proves the claim. $\square$

8. EXACT OPTIMIZATION OF PLANAR PACKETS

For a rotation alphabet, let $z_1, \dots, z_M \in \mathbb{S}^1 \subset \mathbb{C}$ be the distinct product phases of words of length B , ordered cyclically. A common rotation of all phases can be absorbed into the decoder centers. The initial direction in (6.1) therefore disappears. Any probability distribution on these phases is implemented by a distribution on representative words of the same length.

Let $\mathfrak{P}_k$ be the finite set of partitions of the cyclically ordered list into at most k nonempty consecutive arcs. For $p \in \Delta_M$, set

$$(8.1) \quad F_k(p) = \max_{\mathcal{P} \in \mathfrak{P}_k} \sum_{A \in \mathcal{P}} \left| \sum_{i \in A} p_i z_i \right|.$$

Theorem 8.1 (Optimal word laws as a cone program). *For the planar alphabet above,*

$$(8.2) \quad \Gamma_{\mathcal{A}}(k, B) = 1 - \min_{p \in \Delta_M} F_k(p).$$

Equivalently, minimize t subject to $p \in \Delta_M$ and, for every $\mathcal{P} \in \mathfrak{P}_k$,

$$(8.3) \quad \left\| \sum_{i \in A} p_i (\Re z_i, \Im z_i) \right\|_2 \leq r_{\mathcal{P}, A} \quad (A \in \mathcal{P}), \quad \sum_{A \in \mathcal{P}} r_{\mathcal{P}, A} \leq t.$$

This is a finite second-order cone program with attained optimum. A fixed law can be evaluated by cyclic segmentation in $O(kM^3)$ arithmetic operations, including computation of all interval sums. The optimization over laws may have exponentially many partition constraints.

Proof. The decoder centers are on the unit circle; maximizing their inner product is therefore exactly nearest chordal-distance assignment.

For fixed centers $v_1, \dots, v_k$ assign each phase to a nearest center in chordal distance; equivalently maximize $\Re(\bar{v}_j z_i)$. Distinct centers have connected cyclic Voronoi arcs, bounded by the angular midpoints of adjacent centers. Ties can be assigned consecutively; coincident centers can be merged. For the induced partition, the contribution of a block A is at most $|\sum_{i \in A} p_i z_i|$.

Conversely, for any cyclic partition choose the center of each block in the direction of its weighted complex sum, arbitrarily if that sum vanishes. The best assignment to those centers has value at least the sum in (8.1). Thus the maximum over centers equals $F_k(p)$. This proves (8.2); introducing the norm epigraphs proves (8.3). Zero weights follow either directly or by continuity from positive weights.

For a fixed cyclic cut, prefix sums give every block vector in constant time. The usual recurrence for the best j blocks among the first m points checks at most M previous split positions. It costs $O(kM^2)$ for that cut. Repeating for the M cuts costs $O(kM^3)$. This recurrence evaluates a given law; it is not a polynomial-size representation of all constraints in (8.3). Compactness of the simplex and continuity give attainment. $\square$

Nearest-center partitions and vector quantization are classical [5]. The additional optimization here chooses the distribution of actual command words, and Theorem 6.3 turns its value into a constraint on arbitrary hidden registers. There is no assertion that static quantization alone characterizes the least whole-run width.

Proposition 8.2 (A finite exact packet certificate). *For the alphabet consisting of identity and a quarter turn, at word length three the four distinct phases are $1, i, -1, -i$. Their uniform law is optimal and*

$$\Gamma(1, 3) = 1, \quad \Gamma(2, 3) = 1 - \frac{1}{\sqrt{2}}, \quad \Gamma(3, 3) = \frac{2 - \sqrt{2}}{4}, \quad \Gamma(k, 3) = 0 \quad (k \geq 4).$$

Proof. The distortion is concave in the phase law and invariant under its four cyclic rotations. Averaging any law over those rotations proves optimality of the uniform law. For two centers, adjacent pairs give total $\sqrt{2}/2$; a triple and a singleton give only $1/2$. For three centers, one pair and two singletons give $\sqrt{2}/4 + 1/2$. For one center the sum is zero; four centers fit all phases. Formula (8.2) gives the values. A uniform phase law is realized by the four ordinary words with zero, one, two or three initial quarter turns and then idle commands. $\square$

For algebraic rotation matrices, both the general quantifier-elimination formulation and the cone program have algebraic coefficients. No finite list of spherical harmonics is used. For merely computable-real entries, finite numerical approximations are diagnostics unless a rigorous error enclosure is supplied; (6.3) then quantifies how an entry enclosure propagates to a packet certificate.

Corollary 8.3 (Separated phases). *If $M \geq 2k$ distinct length- B word phases have pairwise circular distance at least $s \in (0, 1/2]$, then $\Gamma_{\mathcal{A}}(k, B) \geq s^2$.*

Proof. Use their uniform law. Each open arc of circular radius $s/2$ contains at most one phase. At least half the mass is at distance at least $s/2$ from every one of k centers. On $[0, 1/2]$, $1 - \cos(2\pi t) = 2\sin^2(\pi t) \geq 8t^2$. The average loss is at least s^2 . This is a lower certificate for (8.2), not necessarily its exact value. $\square$

9. ARITHMETIC PROFILES AT EVERY FINITE SCALE

Now $\mathcal{A}_\alpha = \{I, R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_r}\}$, with r fixed. Define

$$(9.1) \quad \psi_\alpha(H) = \min_{0 < \|m\|_1 \leq H} \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}, \quad \eta_\alpha(q) = \max_{1 \leq i \leq r} \|q\alpha_i\|_{\mathbb{R}/\mathbb{Z}}.$$

These are finite functions of the alphabet parameters. The first may vanish in the presence of a rational relation. The upper profile remains usable in that case.

Theorem 9.1 (Finite arithmetic bounds). *Let $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$. For a machine of peak K , put $L = \lceil (2K)^{1/r} \rceil$ and $B = r(L - 1)$. Then*

$$(9.2) \quad \lfloor N/B \rfloor \psi_\alpha(B)^2 \leq \log(1/\kappa).$$

For every integer $q \geq 4$,

$$(9.3) \quad N\eta_\alpha(q) \leq q^2/100 \implies W_{N,0}(\alpha) \leq q.$$

Consequently every finite planar rotation alphabet satisfies

$$(9.4) \quad W_{N,0}(\alpha) \leq \max(4, \lceil \sqrt{50N} \rceil).$$

All upper bounds use common command rows and hold also after adjoining reflection in the first coordinate axis.

Proof. Construct a word of length B in r consecutive segments of $L - 1$ positions. In segment i give J_i copies of the i th rotation followed by idle commands, independently and uniformly for $J_i \in \{0, \dots, L - 1\}$. Differences of count vectors have ℓ_1 norm at most B . If $\psi_\alpha(B) > 0$, the $L^r \geq 2K$ phases are distinct and separated by $\psi_\alpha(B)$. Corollary 8.3 and (6.8), with $-\log(1-x) \geq x$, give (9.2). If the minimum is zero the displayed inequality is automatic. All letters in a segment are actual commands, not a free count input.

For the upper bound let P_q be the regular q -gon inscribed in the unit circle. Choose nearest integers p_i to $q\alpha_i$, and put $\delta_i = 2\pi(\alpha_i - p_i/q)$, so $|\delta_i| \leq \pi/q$. Elementary chord geometry gives

$$(9.5) \quad R_{2\pi\alpha_i}P_q \subseteq \Lambda_i P_q, \quad \Lambda_i = \frac{\cos(\pi/q - |\delta_i|)}{\cos(\pi/q)}, \quad \log \Lambda_i \leq \frac{4\pi^2\eta_\alpha(q)}{q^2}.$$

Indeed the ray of a rotated vertex meets a neighboring edge at radial coordinate $\cos(\pi/q)/\cos(\pi/q - |\delta_i|)$. Integrating the derivative of $\log \cos(\pi/q - t)$ bounds the logarithm by $|\delta_i| \tan(\pi/q)$, and $\tan(\pi/q) \leq 2\pi/q$ for $q \geq 4$.

Take $\Lambda = \max_i \Lambda_i$, including one for the identity. Under (9.3), $\Lambda^N \leq e^{\pi^2/25} < 2$. Since P_q contains the half-unit disk, Theorem 6.5 with $a = 1/2$ applies: $\rho\Lambda^N \leq 1/5 < 1/2$. Explicitly, initialize at mean $x/2$, represent $\Lambda^{-1}R_{2\pi\alpha_i}v_s$ by the vertices for each row, and decode by $2\rho\Lambda^N \langle e_j, v_s \rangle$. This verifies every numerical word, including any deterministic remainder of a converse packet decomposition.

There are $(r + 1)q$ common command rows and two decoder columns, each row having at most three successors. These counts exclude row-description and arithmetic costs. Since $\eta_\alpha(q) \leq 1/2$, (9.4) follows. Reflection preserves P_q ; its contracted image uses the same vertex set. The reflected lower bound is obtained simply by restricting to rotation-only words. $\square$

The pair $(\psi_\alpha, \eta_\alpha)$ is a specialization of the word-distortion and enclosure profiles. It is not assumed to characterize exact finite width. In particular, (9.2) can be weak when long word packets have clusters; the optimized profile of Section 8 is available without that separation relaxation.

Lemma 9.2 (Denominators under a general power bound). *Suppose for $\tau \geq r$ and $b > 0$,*

$$(9.6) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b\|m\|_1^{-\tau} \quad (m \neq 0).$$

For every integer $Q \geq 2$ there is q with

$$(9.7) \quad cQ^{r/(\tau+1-r)} \leq q \leq Q^r, \quad \eta_\alpha(q) \leq Q^{-1}, \quad c = (b/r^{\tau+1})^{r/(\tau+1-r)}.$$

Proof. The $Q^r + 1$ points $j\alpha$ in Q^r equal torus boxes give integers p_i, q with $1 \leq q \leq Q^r$ and $|q\alpha_i - p_i| \leq Q^{-1}$. For $M = \lfloor q^{1/r} \rfloor$, the $(M+1)^r > q$ integer vectors in $\{0, \dots, M\}^r$ have a residue collision for $m \cdot p \pmod{q}$. Subtracting gives $m \neq 0$ with $H = \|m\|_1 \leq rq^{1/r}$ and $q \mid m \cdot p$. Thus

$$bH^{-\tau} \leq \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq H/(qQ), \quad bqQ \leq r^{\tau+1} q^{(\tau+1)/r}.$$

Rearranging yields (9.7). This direct residue argument supplies exactly the quantitative transference needed here, without an unspecified transference constant. $\square$

Corollary 9.3 (Power bounds at finite Diophantine type). *Under (9.6), for constants depending on the fixed data,*

$$(9.8) \quad c_1 N^{r/(2\tau+1)} \leq W_{N,\varepsilon}(\alpha) \leq C_1 N^{\min\{1/2, r(\tau+1-r)/(\tau+1+r)\}}.$$

No matching claim is made when $\tau > r$.

Proof. From (9.2), $N < B + b^{-2} B^{2\tau+1} \log(1/\kappa)$, while $B \leq r(2K)^{1/r}$. This gives the lower power after adjusting finitely many initial horizons. Lemma 9.2 bounds $N\eta_\alpha(q)/q^2 \leq c^{-2} NQ^{-1-2r/(\tau+1-r)}$. Choose Q at constant times $N^{(\tau+1-r)/(\tau+1+r)}$, large enough to satisfy (9.3) and $q \geq 4$. Then $q \leq Q^r$ proves the second candidate upper power; (9.4) proves the first. $\square$

Corollary 9.4 (The retained badly approximable law). *If (9.6) holds with $\tau = r$, then $W_{N,\varepsilon}(\alpha) = \Theta(N^{r/(2r+1)})$, also with the reflection extension. This is the previously established matched law [8].*

Proof. Substitute $\tau = r$ in (9.8); the two powers coincide. $\square$

For $1, \alpha_1, \dots, \alpha_r$ rationally independent, every optimized finite-block action gap on the circle is zero. Indeed simultaneous approximation supplies unbounded frequencies n with every $n\alpha_i$ close to an integer. Each fixed rotation word then acts almost identically on its Fourier mode. For the reflected alphabet use the real cosine mode, which reflection fixes. This is the established almost-invariance argument, not a claim that the finite packet distortion vanishes.

10. MINIMAL ORDINARY TYPE AND ALMOST-EVERYWHERE WIDTHS

Uniform bad approximation is substantially stronger than the condition needed to determine the logarithmic width exponent. We first deduce the minimal-type assertion without any measure-theoretic hypothesis.

Proof of (1.4) in Theorem 1.1. If $\omega^*(\alpha) = r$, then for every $h > 0$ there is $b_h > 0$ such that

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b_h \|m\|_1^{-r-h} \quad (m \neq 0).$$

Indeed the definition of the limsup gives this inequality for all sufficiently large $\|m\|_1$, and rational independence makes the finite remaining minimum positive. Apply Corollary 9.3 with $\tau = r + h$. It gives

$$\frac{r}{2r+2h+1} \leq \liminf_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \limsup_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \min \left\{ \frac{1}{2}, \frac{r(1+h)}{2r+1+h} \right\}.$$

For $0 < h < 1/(2r-1)$ the second branch is smaller. Let $h \downarrow 0$. This proves (1.4). No positive uniform constant at exponent r was used. $\square$

For completeness we prove a more quantitative metric statement, using only the summable part of metric Diophantine approximation. It does not claim a new metric theorem.

Lemma 10.1 (A summable logarithmic separation). *For Lebesgue-almost every $\alpha \in [0, 1]^r$, and every $s > 1$, there is $b_s > 0$ such that*

$$(10.1) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{b_s}{\|m\|_1^r (\log(2 + \|m\|_1))^s} \quad (m \neq 0).$$

Such a vector has $\omega^(\alpha) = r$.*

Proof. For a nonzero integer vector m , the map $\alpha \mapsto m \cdot \alpha \pmod{1}$ sends Haar measure on the torus to uniform circle measure. There are at most $C_r H^{r-1}$ integer vectors with $\|m\|_1 = H$. Hence the measures of the events

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} < [\|m\|_1^r (\log(2 + \|m\|_1))^s]^{-1}$$

have finite sum, bounded by a constant times $\sum_{H \geq 1} [H(\log(2 + H))^s]^{-1}$. The measure of their tail union tends to zero. Almost every vector violates the unscaled inequality only finitely often. Remove the measure-zero rational hyperplanes and reduce the constant to cover the finite exceptions. Intersect the full-measure sets for rational $s > 1$. For a real $s > 1$, choose a rational $1 < s_0 < s$; its stronger bound implies the bound with s , after a harmless change of constant. This gives the assertion on one full-measure set.

This bound implies $\omega^*(\alpha) \leq r$. For the reverse inequality, place the $(M + 1)^r$ values $m \cdot \alpha \pmod{1}$, $m \in \{0, \dots, M\}^r$, in fewer than $(M + 1)^r$ intervals of equal length. A collision gives a nonzero difference of norm at most rM and distance $O(M^{-r})$. As $M \rightarrow \infty$ the norms of some such differences must tend to infinity, since rational independence excludes an exact relation. Thus $\omega^*(\alpha) \geq r$. $\square$

Proposition 10.2 (Quantitative bounds under logarithmic separation). *Suppose (10.1) holds for some $s > 0$ and $b_s > 0$. Then there are $c, C > 0$ such that*

$$(10.2) \quad cN^{r/(2r+1)}(\log(N+2))^{-2rs/(2r+1)} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)}(\log(N+2))^{2r^2s/(2r+1)}.$$

Proof. The packet bound (9.2) gives

$$N < B + b_s^{-2} B^{2r+1} (\log(2 + B))^{2s} \log(1/\kappa).$$

With $B \leq r(2K)^{1/r}$ this implies $N \leq C_0 K^{(2r+1)/r} (\log(2 + K))^{2s}$. If $K > N$, the proposed lower bound is already true after reducing its constant. Otherwise $\log(2 + K) \leq \log(2 + N)$, and rearrangement proves the first inequality of (10.2).

For the upper inequality repeat the residue argument of Lemma 9.2. It produces $q \leq Q^r$, $\eta_\alpha(q) \leq Q^{-1}$ and $H \leq rq^{1/r}$ with

$$\frac{b_s}{H^r (\log(2 + H))^s} \leq \frac{H}{qQ}.$$

Since $q \leq Q^r$, it follows for $Q \geq 2$ that

$$(10.3) \quad q \geq c_0 \frac{Q^r}{(\log(Q+2))^{rs}}.$$

Indeed $b_s q Q \leq H^{r+1} (\log(2 + H))^s \leq C_r q^{1+1/r} (\log(Q+2))^s$, and raising the rearranged inequality to the r th power gives (10.3).

Choose an integer

$$Q = \left\lceil C_2 N^{1/(2r+1)} (\log(N+2))^{2rs/(2r+1)} \right\rceil.$$

For sufficiently large fixed C_2 , (10.3) implies $N\eta_\alpha(q)/q^2 \leq 1/100$ and $q \geq 4$. To see the constant choice explicitly, for large N one has $\log(Q+2) \leq C_3 \log(N+2)$ with C_3 fixed; thus the ratio is bounded by a constant times $C_2^{-(2r+1)}$. Enlarge the final constant for the finitely many initial horizons. Theorem 9.1 gives an exact machine with $q \leq Q^r$ labels, proving the upper bound. $\square$

Completion of Theorem 1.1. Apply Lemma 10.1 with $s = 1 + \delta$ and then Proposition 10.2. This yields (1.5) on one full-measure set for all fixed $\delta > 0$. These statements concern a fixed alphabet before the horizon grows. None of their constants is asserted uniform in r or near a rational relation. $\square$

10.1. Rational dependencies without redundant exponents. The number of listed rotations need not be the relevant arithmetic rank. The following result covers finite rational reparametrizations rather than assigning one exponent to every redundant letter.

Corollary 10.3 (Finite rational extensions). *Suppose an alphabet contains I and rotations through angles $\beta_1, \dots, \beta_d$, where $1, \beta_1, \dots, \beta_d$ are rationally independent and $\omega^*(\beta) = d$. Suppose every additional angle belongs to $\mathbb{Q} + \sum_{j=1}^d \mathbb{Q}\beta_j$. Then*

$$\lim_N \frac{\log W_{N,\varepsilon}}{\log N} = \frac{d}{2d+1}.$$

The same assertion holds on a full-measure set of β with the logarithmic upper and lower powers of (1.5), replacing r by d . Reflection may be added. If all angles are rational, exact width is bounded independently of N .

Proof. Restriction to the displayed d -angle subalphabet gives the lower bound. Choose one common denominator h and integers t_i, n_{ij} so that every angle is $t_i/h + \sum_j n_{ij}\gamma_j$ with $\gamma_j = \beta_j/h$. Multiplication by h shows $\|m \cdot \gamma\|_{\mathbb{R}/\mathbb{Z}} \geq h^{-1} \|m \cdot \beta\|_{\mathbb{R}/\mathbb{Z}}$. Thus every separating power or logarithmic inequality for β holds for γ with a changed constant.

Take a simultaneous approximant p_j/q to γ from the proof above. A command's irrational part has residual angle at most $2\pi(\sum_j |n_{ij}|)/(qQ)$. The same polygon construction therefore works, after changing only fixed constants. Store in addition the rational phase $c \in \mathbb{Z}/h\mathbb{Z}$, updated by $c \mapsto c + t_i$. The decoder rotates the stored polygon vector by $2\pi c/h$. The peak, including the held answer, is $\max(2, hq)$, not an uncharged phase. Reflection sends c to $-c$ and conjugates the polygon phase; the regular polygon and the construction are reflection-invariant. The exact expectation identity is unchanged. The finite multiplier h does not change a logarithmic exponent or the displayed logarithmic powers.

For a purely rational alphabet, choose a common period h and retain the seed identifier and the accumulated phase. This uses at most $4h$ command labels for our four planar seeds, plus the held binary answer convention. No infinite-gap hypothesis is involved. $\square$

11. EVERY IRRATIONAL ANGLE HAS CUBIC SCALES

We keep the two-letter alphabet $\{I, R_{2\pi\alpha}\}$ and the planar resource and error conventions. No Diophantine lower bound is assumed in this section. The needed continued-fraction estimates are classical [3]; we include their short proof to specify the exact finite inequalities being used.

Lemma 11.1 (Convergent separation). *Let p_n/q_n be the convergents of an irrational α , ignoring the finitely many initial indices with repeated denominator. Then*

$$(11.1) \quad \frac{1}{q_{n+1} + q_n} < |q_n\alpha - p_n| < \frac{1}{q_{n+1}}, \quad \min_{1 \leq l < q_n} \|l\alpha\|_{\mathbb{R}/\mathbb{Z}} > \frac{1}{q_n + q_{n-1}} > \frac{1}{2q_n}.$$

Proof. The convergent recurrences imply $p_n q_{n-1} - p_{n-1} q_n = \pm 1$ and alternating signs of $e_n = q_n \alpha - p_n$. If ξ is the next complete quotient, then

$$\alpha = \frac{p_n \xi + p_{n-1}}{q_n \xi + q_{n-1}}, \quad a_{n+1} < \xi < a_{n+1} + 1.$$

It follows that $|e_n| = 1/(q_n \xi + q_{n-1})$, proving the first pair.

For $0 < l < q_n$ and an integer p , the determinant identity writes $(p, l) = a(p_n, q_n) + b(p_{n-1}, q_{n-1})$ with integers a, b . Here $b \neq 0$; a, b cannot both be positive, and cannot both be negative. If $a = 0$ the error has modulus at least $|e_{n-1}|$. Otherwise they have opposite signs, so ae_n and be_{n-1} have the same sign and again $|l\alpha - p| \geq |e_{n-1}|$. Minimize over p and apply the first pair with index $n - 1$. This proves the remaining inequalities. $\square$

Theorem 11.2 (Universal cubic subsequences). *Set $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$, $C_\kappa = 8(1 + \log(1/\kappa))$, and*

$$M_n = \lceil C_\kappa q_n^3 \rceil, \quad m_\kappa = \lceil 100(C_\kappa + 1) \rceil.$$

For all sufficiently large n ,

$$(11.2) \quad q_n/2 < W_{M_n, \varepsilon}(\alpha) \leq m_\kappa q_n.$$

The upper bound is exact, uses common command rows and requires no regularity of the partial quotients.

Proof. Use words of length $B = q_n - 1$ whose number of initial active letters ranges uniformly over $0, \dots, q_n - 1$. Lemma 11.1 gives q_n distinct phases with separation greater than $1/(2q_n)$. If the proposed peak K were at most $q_n/2$, Corollary 8.3 and the packet budget would give

$$\left\lfloor \frac{M_n}{q_n - 1} \right\rfloor \frac{1}{4q_n^2} \leq \log(1/\kappa).$$

The left side is at least $C_\kappa/4 - 1/(4q_n^2)$, strictly greater than the right side. This proves the lower bound against all hidden-state machines.

For the upper bound use the polygon denominator $Q_n = m_\kappa q_n$. Distance to the nearest integer is subadditive, so

$$\|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq m_\kappa |q_n \alpha - p_n| < m_\kappa / q_{n+1} \leq m_\kappa / q_n.$$

Consequently $Q_n^2 / [100 \|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}}] > m_\kappa q_n^3 / 100 \geq (C_\kappa + 1) q_n^3 \geq M_n$. Equation (9.3) gives the exact upper construction. The constants have been chosen for transparency, not optimality. $\square$

Proposition 11.3 (Resonant long horizons). *For any irrational α and integer $q \geq 4$ with $N_q = \lfloor q^2 / (100 \|q\alpha\|_{\mathbb{R}/\mathbb{Z}}) \rfloor \geq 1$, one has $W_{N_q, 0}(\alpha) \leq q$. If α is Liouville, there is a sequence $N \rightarrow \infty$ on which $\log W_{N, \varepsilon}(\alpha) / \log N \rightarrow 0$.*

Proof. The first assertion is the finite enclosure implication (9.3). For a Liouville number choose arbitrarily large A and unbounded denominators q with $0 < \|q\alpha\|_{\mathbb{R}/\mathbb{Z}} < q^{-A}$. The first assertion then supplies horizons $N_q \geq q^{A+2}/200$ for large q , with width at most q . The logarithmic ratio is at most $1/(A+2) + o(1)$. A diagonal sequence with $A \rightarrow \infty$ proves the claim, since widths are at least two. $\square$

Proof of Theorem 1.2. Equation (11.2) and $M_n = \Theta(q_n^3)$ give the first assertion and a subsequence of logarithmic exponent $1/3$. Proposition 11.3 gives the zero lower limit for Liouville angles, and (9.4) gives the upper limit at most $1/2$.

It remains to verify unboundedness, which does not follow from a subsequence alone. Fix any proposed width K . The $2K$ phases of counts $0, \dots, 2K - 1$ are distinct for an irrational angle and have some positive minimum separation s_K . Independent words of length $2K - 1$ imply $\lfloor N/(2K - 1) \rfloor s_K^2 \leq \log(1/\kappa)$ for a width- K machine. This fails for all sufficiently large N . Thus $W_{N, \varepsilon} \rightarrow \infty$. $\square$

Corollary 11.4 (An explicit fluctuating alphabet). *Let $\alpha_L = \sum_{j=1}^{\infty} 10^{-j!}$. For $q_m = 10^{m!}$ and all large m , the horizon*

$$N_m = \lfloor q_m^{m+2} / 200 \rfloor$$

admits an exact machine with at most q_m labels. The same fixed two-letter alphabet also has the cubic subsequences of Theorem 11.2; hence no single logarithmic width exponent exists. Adding reflection preserves these conclusions.

Proof. The first m summands have denominator dividing q_m , and their remaining positive tail is less than $2 \cdot 10^{-(m+1)!}$. Thus $0 < \|q_m \alpha_L\|_{\mathbb{R}/\mathbb{Z}} < 2q_m^{-m}$ for large m , proving the horizon bound by (9.3). The same tail estimates preclude rationality and show the Liouville property. The general cubic theorem applies. Reflection leaves the upper polygons invariant, while the rotation-only test words give the same lower bounds. $\square$

These theorems describe different scales of one and the same experiment. A small label count on a long resonant horizon is not a constant-width simulator for all horizons, and it need not persist across the gaps between successive approximants. The exact finite optimum and the Liouville upper logarithmic limit remain separate quantitative questions. None of the proofs substitutes approximate output for the exact upper realizations.

12. FINITE RANDOM BITS AND UNIFORM SPACE

Exact real rows and finite-bit algorithms are different resources. We first give a finite perturbation statement and then remove the row table from a concrete uniform construction. Only this section uses a computational model. It does not change the exact atomic-row conclusions proved elsewhere.

Lemma 12.1 (Dyadic perturbation of a stochastic transducer). *Suppose initialization and every command row have at most s nonzero entries. They can be replaced by denominator- 2^b rows, on the same label sets, with row-TV error at most $(s-1)2^{-b}$. Rounding the terminal binary probabilities adds at most 2^{-b} . Consequently every numerical word changes by at most $((N+1)(s-1)+1)2^{-b}$ in binary total variation. Each rounded row can be sampled with exactly b independent fair bits.*

Proof. Order the s successors, round the first $s-1$ probabilities down to multiples of 2^{-b} , and give the remaining mass to the last successor. The total removed mass is at most $(s-1)2^{-b}$, which is exactly the row total variation. Couple the initializations and then the command updates, or telescope products using TV contraction, to sum their worst-row errors. A terminal Bernoulli parameter can be rounded down with error at most 2^{-b} . A uniform integer in $\{0, \dots, 2^b - 1\}$ selects one of the cumulative dyadic intervals. This uses a fixed number of bits, with no retained random tape or rejection-time side channel. $\square$

For a table implementation the storage is $O(|\mathcal{A}|Ks(\log K + b))$ bits for common rows of width K , in addition to initialization and decoder data. Cut-dependent rows can require a factor N more storage. Lemma 12.1 is the elementary stochastic-rounding principle; it is not claimed as a new finite-bit simulation method. The next construction uses no such table.

Definition 12.2 (Uniform command-stream model). A fixed finite program receives N in binary, one of the four planar seeds, exactly N one-way command symbols, and one final coordinate query. It has a work tape and fresh independent fair bits. The command input cannot be reread. All counters, arithmetic registers, stored parameters, random words and output storage count toward work space. An external source waits while a command is processed; waiting time is not an input or a persistent storage channel. The finite control is fixed independently of N . The program has no horizon-dependent advice or transition table.

An angle has an admissible digit procedure if a fixed program computes a dyadic approximation within 2^{-p} in polynomial time in p and $O(p)$ work bits. The same fixed r is used throughout. This explicit hypothesis excludes a free oracle for an arbitrary noncomputable real. Algebraic angles, in particular the examples below, meet it by rational bisection with fixed-degree integer arithmetic.

Theorem 12.3 (A table-free finite-bit compiler). *Fix r angles with admissible digit procedures, rational $0 < \rho \leq 1/10$, and rational $0 < \varepsilon < 1$. There is a uniform program in Definition 12.2 whose every binary output row has TV error at most ε from the rotation experiment. It uses*

$$(12.1) \quad O_{r,\alpha,\rho}(\log(N+2) + \log(1/\varepsilon)) \text{ work bits}, \quad (N+2)b \text{ fair bits},$$

where $b = \lceil \log_2(16(N+2)/\varepsilon) \rceil$. Its total running time is polynomial in $N + \log(1/\varepsilon)$ for fixed digit procedures. Its clean stochastic register has $q+1$ labels, including one central label, with $q = O(\sqrt{N})$.

If a denominator $q_0 \geq 4$ satisfies $N\eta_\alpha(q_0) \leq q_0^2/2000$, the search may choose $q \leq q_0$. Consequently the clean-register upper exponents and logarithmic bounds of Theorem 1.1 hold for this compiler as well, with changed constants. These clean labels do not include its scratch space or deterministic parameter registers; the full work bound is (12.1).

Proof. We give the parameter search, real reference rows, dyadic rounding, and finite-space evaluation separately.

Parameter search. Put $p = \lceil \log_2(16000(N+2)/\varepsilon) \rceil$ and obtain dyadic $\tilde{\alpha}_i$ with error at most 2^{-p} . For $q = 4, 5, \dots$ compute, by integer arithmetic,

$$e(q) = \max_i \|q\tilde{\alpha}_i\|_{\mathbb{R}/\mathbb{Z}} + q2^{-p}.$$

Stop at the first q for which $Ne(q) \leq q^2/1000$. The test is rational and certifies both $\eta_\alpha(q)$ and $\eta_{\tilde{\alpha}}(q)$ by the Lipschitz property of distance to $\mathbb{Z}$. It succeeds by $q = \max(4, \lceil \sqrt{2000(N+2)} \rceil)$: $\|q\tilde{\alpha}_i\| \leq 1/2$ and the added term is smaller than one at this bound. If q_0 satisfies the stronger displayed condition, then $e(q_0) \leq \eta_\alpha(q_0) + 2q_02^{-p}$ and $2Nq_02^{-p} \leq q_0^2/2000$. Hence that candidate succeeds. No exact comparison of an irrational angle with a rational sector boundary is performed.

Reference rows with slack. Use the regular q -gon $v_j = (\cos(2\pi j/q), \sin(2\pi j/q))$ and one additional label $*$ representing zero. Set $\Lambda = 1 + 1/(4N)$. For $N \geq 1$ the dilation estimate (9.5) and the search condition give

$$\log \Lambda \geq 1/(5N) > 4\pi^2\eta_{\tilde{\alpha}}(q)/q^2.$$

Thus $\Lambda^{-1}R_{2\pi\tilde{\alpha}_i}v_j$ lies strictly inside the polygon. The case $N = 0$ is handled by direct rounding of the seed response and a fixed amount of space. Write $q\tilde{\alpha}_i = n_i + t_i$, with integer n_i and $0 \leq t_i < 1$, and put $h = 2\pi/q$. The command row has successors $j + n_i$, $j + n_i + 1$ modulo q , and $*$, with respective weights

$$(12.2) \quad A_i = \frac{\sin((1-t_i)h)}{\Lambda \sin h}, \quad B_i = \frac{\sin(t_i h)}{\Lambda \sin h}, \quad 1 - A_i - B_i.$$

They are nonnegative by the dilation bound, and their vector mean is the required contracted rotation. The row at $*$ stays there. Initialize with vector mean $x/2$ using the same chord formula with radial factor $1/2$ and the exact seed angle $0, 1/4, 1/2, 3/4$. Decode with mean $2\rho\Lambda^N \langle e_j, v_s \rangle$, or zero at $*$. Since $\Lambda^N \leq e^{1/4}$, all decoder means are strictly inside $[-1, 1]$. The usual mean induction realizes every word for $\tilde{\alpha}$ exactly. These are reference real rows only.

Certified finite rows. Compute outward intervals for the first two weights in each row, each of width at most 2^{-b} . Clip their lower endpoints at zero and round down to denominator 2^b ; assign the residue to $*$. Relative to the reference row, at most $4 \cdot 2^{-b}$ mass is moved. Evaluate and round each terminal Bernoulli parameter within $2 \cdot 2^{-b}$. TV contraction therefore bounds initialization, N updates and final decoding by $(4N+6)2^{-b}$. Replacing the true angles by their dyadic approximants changes a product rotation by at most $2\pi N 2^{-p}$ in operator norm, and changes a target binary row by at most $\pi\rho N 2^{-p}$. The sum is less than ε with the chosen p, b . Use exactly b fair bits at each draw, including dummy bits when the central row is deterministic. The random-bit count is $(N+2)b$.

Work and uniform evaluation. Only q , the current vertex or central flag, the horizon and command counter, the r dyadic angles, and $O(r)$ offsets and dyadic weights are retained. The latter specify circulant rows, not a q -row table. The denominator search uses integers of $O(p + \log q)$ bits. Elementary sine and cosine on bounded intervals can be evaluated by fixed-point Taylor recurrences with outward rounding; π can be evaluated by Machin's arctangent identity. There are $O(b + \log q)$ terms and only a fixed number of live accumulators. Since $\sin(2\pi/q) \geq 4/q$ for $q \geq 4$, division loses only $O(\log q)$ bits. Repeated squaring evaluates Λ^N with $O(\log N)$ stages, all intermediate true values between 1 and $e^{1/4}$. An additional $O(\log(N+2) + \log(b+2))$ guard bits bounds accumulated rounding errors. Interval precision can instead be doubled until the stated widths are certified; these error estimates bound the final precision by the same order. No integer of unbounded exact denominator is stored. Every live register has $O(b + p + \log N)$ bits, and the number of registers is fixed for fixed r . This proves the work bound and a polynomial time bound, including parameter search. Finally the finite arithmetic upper proofs remain valid with $1/100$ replaced by $1/2000$, since only fixed constants change. Their candidate denominators therefore bound the search. $\square$

The proof describes a fixed-point implementation, not the numerical behavior of a floating-point library. The accompanying integer-interval reference code certifies each rounded row and uses fresh bounded-length random integers; its Python object overhead is not identified with the Turing work-space constant.

Corollary 12.4 (Sharp work-space order for explicit alphabets). *Let $\alpha_i = 2^{i/(r+1)}$, $1 \leq i \leq r$, and fix rational $0 < \varepsilon < \rho/(2\sqrt{2})$ and $0 < \rho \leq 1/10$. In Definition 12.2, numerical simulation requires and suffices with $\Theta(\log N)$ work bits. More precisely every fixed program using $s(N)$ work bits obeys*

$$(12.3) \quad s(N) \geq \frac{r}{2r+1} \log_2 N - O(\log \log(N+2)).$$

The compiler's clean register has $O(N^{r/(2r+1)})$ labels. This is not a claim that all its internal fair-bit states retain that label exponent or that the leading coefficient of work space is optimal.

Proof. Put $\theta = 2^{1/(r+1)}$. For a nonzero integer vector m with $H = \|m\|_1 \geq 1$, choose the integer p nearest to $-\sum_i m_i \theta^i$. The algebraic integer $P(\theta) = p + \sum_i m_i \theta^i$ is nonzero: the minimal polynomial $t^{r+1} - 2$ is Eisenstein and has degree $r+1$. Each of its other r conjugates has modulus at most $(9/2)H$, since $|p| \leq 2H + 1/2$ and $|\theta^i| < 2$. Its nonzero integer norm gives

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq (2/9)^r H^{-r}.$$

This classical algebraic example is credited in [4]; it is not new arithmetic. The inherited badly approximable theorem, Corollary 9.4, now supplies the all-hidden lower bound and the matching candidate denominators for the compiler. Bisection for θ and fixed-degree powers meet the digit hypothesis. A fixed program with $s(N)$ work bits has at most $C(s(N) + 1)^C 2^{s(N)}$ command-boundary configurations. Its private randomness since the last boundary is integrated into the next stochastic row, so it induces a legal atomic realization of at most that width. Apply the lower bound $W_{N,\varepsilon} \geq cN^{r/(2r+1)}$ and take logarithms. The polynomial factor accounts for the fixed number of work-tape head positions; it contributes $O(\log \log N)$ unless the displayed lower bound is already automatic. Theorem 12.3 supplies the upper work bound. $\square$

13. REALIZATION THEORY, ARITHMETIC AND COMPUTATIONAL SEMANTICS

Positive order and cone invariance. In the classical stationary problem one seeks a fixed nonnegative state-space realization of a transfer function. Minimal signed order need not equal minimal positive order. Benvenuti–Farina's Theorems 2–3 formulate positive existence by an invariant polyhedral

cone, either in a minimal linear realization or in a response-sequence space [12]; the 2022 survey concerns positive minimality [13]. Our enclosure construction is such a positive-cone construction after homogenization. The competing machines are different: their rows may vary with the cut and their entire realization may be redesigned for N . Proposition 4.1 makes the corresponding finite controlled normal form explicit. Classical stationary order cannot be substituted for its least simultaneous profile.

Hankel rank and signed weighted automata. For fixed seed and query the mean series is $w \mapsto \rho\langle e_j, U_w x \rangle$, with a signed linear representation of dimension at most D . Adding the constant probability offset requires at most $D + 1$ dimensions. Balle–Panangaden–Precup’s Theorem 2 identifies minimal weighted-automaton dimension with infinite Hankel rank [14]. That is the signed, all-length invariant, not stochastic label cardinality. Their Theorem 11 bounds an ℓ^2 approximation of absolutely summable rational series. Here the unweighted all-word probability series is not summable, and the objective is the maximum numerical error on a fixed length. A small weighted mean-square error under a chosen word law does not control rare-word conditional error. For an explicitly unweighted finite table, Frobenius error does bound entrywise error, but its optimizer still need not be stochastic or compatible. Equation (4.5) exhibits the precise failure of separate ranks to detect the present growing resource.

Stochastic and hidden Markov realization. Finesso–Grassi–Spreij use pseudo-Hankel nonnegative factorizations and an informational-divergence criterion to approximate a stationary output process at prescribed hidden order [15, Sections III–IV]. Their second realization step is itself evidence that a static factorization is not the complete dynamic specification. Our alphabet is externally controlled rather than generated as an observation process. Seed resets and selected query columns are separately specified; worst-word conditional probabilities replace a divergence between output laws. These are explicit changes of datum and objective, not a claim that Hankel positivity was absent from HMM theory.

Ohta’s reachable and null-space reductions, Theorems 6–9 and Remark 10, identify tensor and generalized-Hankel dimensions in the effective linear space of an HMM [16]. We use the same need to distinguish observable rank from hidden state count, but do not infer a positive stochastic basis from a signed quotient. His Section 4.4 also discusses rotational examples descended from earlier HMM realization work. Rotation-based rank/positive-order separations are therefore not claimed new here. Our quantitative theorems concern all horizon-specific stochastic rows and preserve bounded numerical responses on every controlled word.

Kronecker quasi-uniformity. Dick–Goda–Larcher–Pillichshammer–Suzuki [4], Theorem 3.9, characterize quasi-uniformity of the multidimensional sequence $n\alpha \pmod{\mathbb{Z}^r}$ by the simultaneous bound $\eta_\alpha(q) \geq cq^{-1/r}$. Their Lemma 3.11 bounds ratios of successive best simultaneous denominators; Section 3.3 also contains the vector $(2^{1/(r+1)}, \dots, 2^{r/(r+1)})$. These are direct arithmetic antecedents, not new examples of this paper.

Our packet points instead have the form $m \cdot \alpha \pmod{1}$. The relevant separation is dual. The residue argument of Lemma 9.2, with an arbitrary nearest-integer numerator and $\eta_\alpha(q)$ in place of Q^{-1} , proves directly that a dual bound with exponent r implies

$$\eta_\alpha(q) \geq br^{-r-1}q^{-1/r}.$$

Thus the cited quasi-uniformity theorem applies to the uniform badly approximable class used in the predecessor. Its denominator control supplies the arithmetic scales for a polygon upper bound. It supplies neither a compatible stochastic transition table nor a lower bound against discarded hidden history. The logarithmic separation of Lemma 10.1 is weaker than a uniform badly approximable inequality; it is proved by a summable tail estimate, not asserted to imply quasi-uniformity.

What is dynamic in the paired profiles. The geometry of nearest-center distortion and convex hull approximation is classical [5, 11]. A static k -center factorization does not specify how its encoder is calculated without retaining discarded inputs. Conversely, a convex codebook does not force arbitrary hidden states to possess coordinates in that codebook. The two implications in Section 6 avoid both identifications: the converse conditions on the actual state, and the upper theorem provides the same legal rows throughout the run. The packing–dilation balance in Corollary 6.7 is a conditional theorem on independently specified finite profiles, not a claimed equality of static and dynamic optima.

The circle program makes one of these profiles exactly finite, whereas the all-harmonic action-gap optimization of the predecessor remains infinite-dimensional. A finite word length can be exhaustively listed; an unbounded harmonic degree cannot be certified by a finite truncation without a tail theorem. No general action-gap algorithm follows from Proposition 6.1.

Branching programs and probability preservation. At fixed horizon our machine is a probabilistic ordered read-once layered program with arbitrary stochastic rows and bounded real output columns. We distinguish width from total size, uniform description and random-bit complexity. Fourier growth for regular, permutation and width-three branching programs concerns different hypotheses and objectives [9, 10, 6]. Circle-walk harmonic damping and its arithmetic behavior are also established subjects [2]. Our converse can use correlated letters inside independently sampled complete packets; it is not a theorem about average error under fair independent letters.

Preserving a strict cutpoint sign is weaker than preserving a numerical probability. The predecessor’s constant-width sign construction contracts the output amplitude with length and hence does not contradict the present width growth. Here signal and tolerance are fixed, one query is selected after the word, and neither joint answers nor anytime correctness is required. The exact arithmetic upper rows are atomic real kernels. Section 12 gives a separate finite-bit approximation theorem; it does not represent those exact rows by a finite number of fair bits. A row with three successors can still have costly or irrational coefficients.

Conclusions. The finite optimized word profile and the compatible enclosure profile retain quantitative information which qualitative gap data discard. They yield a minimal-type exponent and an almost-everywhere logarithmic estimate, and they exhibit fluctuating numerical width for every Liouville single angle. The reflected versions show that these conclusions survive a particular noncommuting finite extension, without asserting a classification of arbitrary noncommuting nongapped alphabets.

Closed-form finite widths beyond the explicitly solved frontier, the Liouville upper logarithmic limit, optimal enclosure shapes and equality in the distortion–dilation comparison remain distinct quantitative problems. The new profile comparison uses no external spectral gap. For fixed computable planar angles, storing all command counts already gives a classical $O(\log N)$ finite-accuracy work-space upper bound. We do not claim novelty for that upper order alone. The compiler proved here additionally maintains an arithmetic-scale clean stochastic register, generates circulant dyadic rows without a large advice table, and explicitly prices every arithmetic and random-bit register. The earlier expansion, calibration and representation theorems retain their original hypotheses in the unchanged supporting article.

Certificates and optimization. The Gram identity of Theorem 5.1 is a finite-length version of classical weighted-series equivalence, with stochastic rows and bounded numerical decoders imposed separately. In contrast to all-length ℓ_2 truncation, the moment sandwich in Theorem 5.2 targets the maximum error over the finite word set and optimizes over legal stochastic machines. Its global minimum is not supplied by an approximate matrix factorization. The rational grid certificate is complete at every fixed positive margin, but exact rational feasibility is different; the normalized nonnegative-rank example makes that distinction explicit. The new certificate framework does not change any earlier arithmetic exponent or imply a uniform solution of the infinite-gap problem.

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